

The quest for CMB B-modes

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Abstract

This paper provides a pedagogical review of the most recent developments in polarization measurements of the Cosmic Microwave Background (CMB).

1 Introduction

This document is meant to provide a review of the current status of B-mode polarization measurements of the Cosmic Microwave Background (CMB), as well as a presentation of the activities done by the Cosmic Microwave Background group in the Department of Physics of the University of Milan.

The search for primordial B-mode fluctuations in the polarization pattern of the Cosmic Microwave Background is one of the most active and promising fields in the domain of observational cosmology. These B-mode fluctuations are the prediction of a family of cosmological models, and their detection would provide an extraordinary tool to understand what happened during the very first stages of the evolution of the Universe. Unfortunately, if primordial B-modes really exist, they have proven to be extremely faint, in fact fainter than any level reached by the measurements done so far.

This article provides a pedagogical introduction to the field of cosmology and Cosmic Microwave Background science, as well as a presentation of the best technology available to perform the kind of measurements described here. We will first present an overview of the basic concepts of

¹But you should be aware that in cosmology LSS can also be used as an acronym for Large Scale Structure!

Table 1: List of abbreviation used in the paper

Acronym	Meaning	Definition
ADC	Analog-to-Digital (converter)	Sect. 5.1.1, page 35
CMB	Cosmic Microwave Background	Sect. 2.3, page 12
DLFOV	Diffraction-limited field of view	Sect. 5.2, page 42
FWHM	Full Width Half Maximum	Sect. 4.5, page 32
GR	General Relativity	Sect. 2.1, page 3
HEMT	High Electron Mobility Transistor	Sect. 5.1.1, page 35
HWP	Half-wave Plate	Sect. 2, page 38
KID	Kinetic Inductance Device	Sect. 5.1.3, page 38
Λ CDM	Λ Cold Dark Matter (model)	Sect. 2.2, page 9
LSS	Last Scattering Surface ¹	Sect. 2.4, page 13
PSD	Power Spectral Density	Sect. 4.6, page 33
SCM	Standard Cosmological Model	Sect. 2, page 2
TES	Transition Edge Sensor	Sect. 5.1.2, page 36

cosmology needed to understand the importance of B-mode CMB polarization (Sect. 2). Then, we will present a few basic concepts needed to understand the technology and the techniques used in CMB experiments (Sect. 4). Then, in Sect. 5 we will present an overall view of the present state of CMB measurements, both in terms of technologies and experiments.

2 The CMB and the inflationary paradigm

In this chapter we provide a short summary of the Standard Cosmological Model (SCM), a framework that describes the evolution of the Universe from an initial singularity (called *Big Bang*) to its present state through an uninterrupted expansion of space-time. Then, we explain how the CMB was formed and what are its properties. Finally, we present the Inflationary paradigm and explain how the CMB can be used to verify the validity of this paradigm. Readers interested in a deeper and more comprehensive treatment of the subject can refer to several books: [94] is a short but very good introduction to the topic, [44] is an incredibly thoughtful book which requires no prior knowledge of advanced mathematics but provides several insights into the quirks of this discipline, and [107] is an advanced

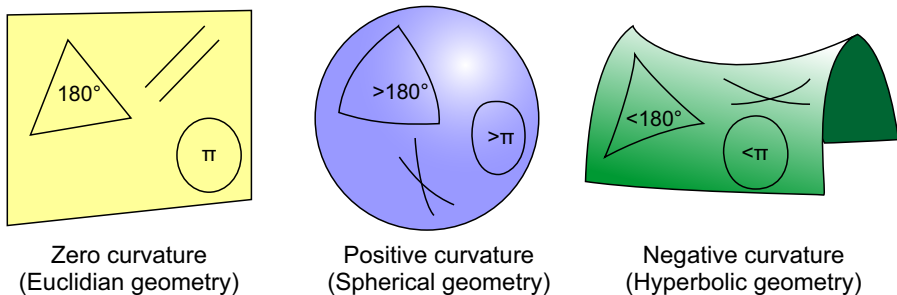


Figure 1: Three kinds of space curvatures. For each curvature, a few basic facts are presented in the vignette: the sum of the internal angles of a triangle, the area of a circle with radius $r = 1$, and the behavior of parallel lines. Euclidean geometries satisfy the postulates and theorems found in Euclid’s “Elements”. Parallel lines can intersect in spherical geometries, and non-parallel lines can be non-intersecting in hyperbolic geometries. According to General Relativity, the local properties of space near some point P belong to one of these three kinds of geometries, depending on the distribution of mass and energy in the neighborhood of P .

textbooks which contains a rigorous presentation of cosmology.

2.1 Friedmann’s equations

It is amazing to think that Einstein’s general relativity (GR), the basic tool used by modern cosmology, was born in 1915, years before we realized that the Universe is far larger than our Milky Way (Hubble 1923; see also Hoskin 1976 for a review). Einstein’s main motivation for building GR was not to develop a new cosmological model, but to solve the puzzle of the unexplained perfect match between gravitational and inertial masses in Newton’s dynamics, which he perceived as a logic flaw. Nevertheless, physicists were keen to start using GR for cosmology just after Eddington confirmed the validity of the theory with his famous observation of the 1917 Solar Eclipse [57]. That same year, Einstein published his own cosmological static model, where he introduced the now famous Λ term as a way to counterbalance gravity’s attraction; this had the purpose to keep the space-time fabric of the Universe the same as time passes [57]. Einstein’s model used the assumption that the universe be *homogeneous* (its properties are the same at any point) and *isotropic* (the Universe looks

the same along any direction); these assumptions are generally believed to be valid even today. A few years later (1920), de Sitter published a paper which modeled an expanding universe and predicted an observable spectral redshift of distant objects. In 1922 Friedmann proposed his family of models for an homogeneous and isotropic expanding universe, according to the following space-time metric² (in spherical coordinates, using the signature $+- --$):

$$ds^2 = c^2 dt^2 - a^2(t) (dr^2 + \tilde{r}^2(d\theta^2 + \sin^2 \theta d\phi^2)), \quad (1)$$

where a is the so-called *scale factor*, $\tilde{r}$ can either be $R_C \sinh r/R_C$ (negative curvature with radius R_C), r (zero curvature), $R_C \sin r/R_C$ (positive curvature with radius R_C). The meaning of the curvature is explained in Fig. 1. The value of the scale factor a at some time t is a pure number that multiplies the spatial part of the metric, and therefore it quantifies how space dilates or contracts; in a classical, Newtonian universe, $a(t) = 1 \forall t$. The functional form of a can be calculated from Einstein's field equations of General Relativity:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}, \quad (2)$$

where $R_{\mu\nu}$ is Ricci's tensor, R is Ricci's scalar, $g_{\mu\nu}$ is the metric tensor, Λ is the cosmological constant (not present in Friedmann's seminal work, but now believed to be non-zero in order to account for present-day observational data), and $T_{\mu\nu}$ is the stress-energy tensor, which describes the distribution of mass-energy in space-time. The terms $R_{\mu\nu}$, R , and $g_{\mu\nu}$ can be calculated using the metric in Eq. (1), while $T_{\mu\nu}$ and Λ need to be measured through observations.

Solving Einstein's Eq. (2) with the metric in Eq. (1) leads to the so-called *Friedmann equations*, which describe the behavior of the scale factor a in terms of a few observables:

$$\left(\frac{\dot{a}(t)}{a(t)}\right)^2 = -\frac{kc^2}{a^2(t)} + \frac{8\pi G}{3c^2}\epsilon(t) + \frac{\Lambda}{3}, \quad (3)$$

$$\frac{\ddot{a}(t)}{a(t)} = -\frac{4\pi G}{3c^2}(\epsilon(t) + 3P(t)) + \frac{\Lambda}{3}, \quad (4)$$

²Unlike most of the textbooks in General Relativity, this paper does not use the convention $G = c = \hbar = 1$, in order to keep all measure units explicit.

where ϵ is the energy density of the Universe (supposed to be the same everywhere, because of the assumption of an homogeneous and isotropic space), P is the pressure (ditto), and k is the curvature parameter. Nowadays there is strong observational evidence that the Universe has nearly zero curvature, and several models assume $k = 0$ (the so-called³ *flat Universe hypothesis*). In order to make these equations solvable, a *fluid equation* to establish a relationship between ϵ and P is needed:

$$\dot{\epsilon}(t) + 3\frac{\dot{a}(t)}{a(t)}(\epsilon(t) + P(t)) = 0, \quad (5)$$

as well as suitable initial conditions; these must include one condition on $a(t)$ and one on $\dot{a}(t)$. Usually, the initial conditions are set at $t = t_{\text{now}}$ and are thus $a(t_{\text{now}}) = 1$, $\dot{a}(t_{\text{now}}) = H_0$, with H_0 the Hubble constant (see below).

In order to link the model described by Friedmann equations 3 and 4 with observations, cosmologists have defined a few important quantities derived from $a(t)$. The quantity

$$H(t) = \frac{\dot{a}(t)}{a(t)} \quad (6)$$

is called the *Hubble parameter*, and its value at time $t = t_{\text{now}}$ is the so-called *Hubble constant*, roughly equal to 70 Km/s/Mpc. Note that the measure unit of $H(t)$ is the inverse of a time: the physical significance of the *Hubble time* $1/H(t)$ is the order of magnitude of the time required for a light ray to propagate over the distance $c/H(t)$ (called *Hubble length*), which in turn provides a rough estimate for the size of the observable Universe⁴ at time t . Another important quantity is the *cosmological redshift*, which quantifies the stretch in the wavelength of a photon due to the dilation of space caused by the scale factor a . If the photon's wavelength at the time of emission was λ_{em} , and the wavelength at the time of the observation is λ_{obs} , and if the only reason why the wavelength changes is because of the

³The term “flat” is rather infortunate, as it seems to suggest that the Universe is bidimensional; we prefer the term *Euclidean*, which is less ambiguous.

⁴The observable Universe is a sphere within the Universe, centered on some observer, with a radius R large enough to contain all the points of space which have been visible by the observer at some time in the past. For a static, non-expanding Universe of finite age t_{age} , $R = ct_{\text{age}}$.

cosmological expansion, then the redshift is

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{em}}}{\lambda_{\text{em}}}, \quad \Rightarrow \quad z(t) = \frac{1}{a(t)}, \quad (7)$$

where the last equality follows from Eq. (1) and is true when considering cosmological timescales (i.e., timescales much larger than λ_{em}/c). Redshifts are widely used as a measure of time: a is believed to be a monotonically increasing function of t , and therefore $z = z(t)$ can be inverted so that there is a unique correspondence between any time t and its redshift. Of course, to know how to calculate this inversion, one needs to have a good estimate of $a(t)$, which in turns requires a good knowledge of ϵ , P , and Λ (see Eqs. 3 and 4).

There is one last important quantity that can be derived from the Friedmann equations (3) and (4), and it is related to the average energy density of the Universe. In the hypothesis of a Euclidean universe with $k = 0$, Eq. (3) reduces to

$$H^2(t) = \frac{8\pi G}{3c^2} \epsilon(t), \quad (8)$$

which means that for any value of the Hubble parameter $H(t)$ there is a *critical density*

$$\epsilon_c(t) = \frac{3c^2}{8\pi G} H^2(t), \quad (9)$$

which is the energy density that a Euclidean Universe must possess: larger densities lead to a positively curved space, smaller densities to a negatively curved space. The density of the Universe today is very close to the critical density [81], and it is roughly one proton per 200 liters. It is customary to normalize energy densities using $\epsilon_c(t)$, in order to make use of pure numbers:

$$\Omega(t) = \frac{\epsilon(t)}{\epsilon_c(t)}. \quad (10)$$

The use of Ω symbols is widespread in cosmology: the most used conventions express the density of baryons as $\Omega_b = \epsilon_b/\epsilon_c$, and similarly for Ω_Λ (dark energy density), Ω_{dm} (dark matter density), and so on. The Friedmann equations can also be rewritten using Ω ; for instance, the first equation becomes

$$1 - \Omega(t) = -\frac{kc^2}{a^2(t)H^2(t)} \quad (11)$$

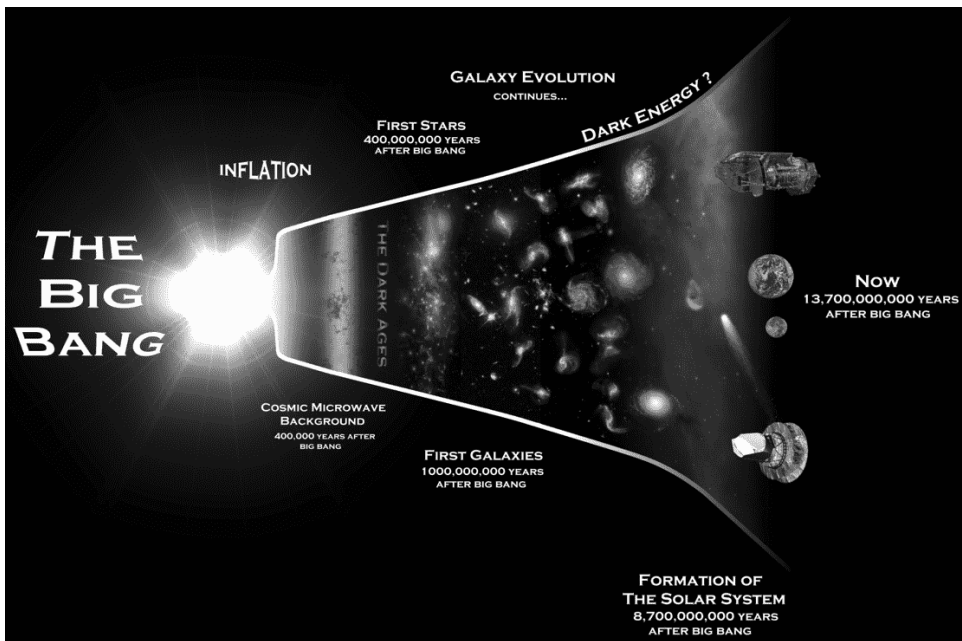


Figure 2: Depiction of the evolution of the Universe according to the Big Bang model. Time t flows from left to right. The white lines which borders the plot is a rough depiction of the behavior of $a(t)$, the solution of Friedmann's equation. Taken from <http://planck.cf.ac.uk/science/timeline/universe>.

This equations makes evident the importance of $\Omega(t_{\text{now}})$ as a way to characterize the geometry of the Universe: as we mentioned above, this parameter is so close to one that it is usually assumed that the Universe we live in is Euclidean.

2.2 Our present understanding of the history of the Universe

Figure 2 depicts the evolution of the Universe according to the inflationary paradigm and the Big Bang theory. Following the figure from left to right, we see that at $t = 0$, immediately after the Big Bang, the global fabric of the Universe's space-time underwent a rapid expansion called *inflation*. This initial inflation, lasting some 10^{-33} s or so, enlarged a very small region of space to macroscopic scales and flattened out dishomogeneities in the primordial plasma, and happened at some time within the first 10^{-12} s

since the Big Bang. We will discuss inflation in more detail in Sect. 3.1, as the search for experimental evidence of the existence of this phase are the main topic of this article.

After the end of the inflationary phase, the expansion proceeded more regularly. The progressive expansion of space-time caused the temperature of the plasma to drop steadily, and the reduced kinetic energy of its particles allowed them to combine and form heavier particles. By $t = 10$ s, protons and neutrons were already in place, and Helium was present⁵ with the mass ratio He/H already set to the value of 25% that we still observe today, though slightly increased by stellar nucleosynthesis. At this time, the main constituents of the Universe were non-interacting dark matter, baryons, electrons, and photons.

The high temperatures in the Universe allowed matter and radiation to be in thermal equilibrium thanks to Thomson scattering between photons and electrons. This means that the thermodynamic temperature of the three species was the same: $T_p(t) = T_e(t) = T_\gamma(t)$, and one can to speak generally⁶ of the “temperature of the Universe” with no further specification. This phase ended when the temperature was low enough ($T \approx 3000$ K) to allow the formation of neutral atoms from the combination of nuclei and electrons. This broke the equilibrium between matter and radiation, causing them to lose their ability to keep their mutual equilibrium. What happened is that the number density of electrons n_e went to zero⁷ with some rate, and the mean free path λ_γ of photons diverged⁸ to infinity. The *decoupling* of matter and radiation is conventionally considered to be the time t_{dec} when the mean free path became larger than the scale of the observable Universe, i.e., $\lambda_\gamma = c/H(t_{\text{dec}})$. This happened at $t_{\text{dec}} \approx 380\,000$ yr, and primordial photons have been freely to propagate through space with little to no interaction with matter since then. This event is marked in Fig. 2 with the label “Cosmic Microwave Background”.

After decoupling, dishomogeneities in the matter distribution continued to grow until the first stars and galaxies formed at $z \sim 10 \div 20$ (be-

⁵Traces of light elements up to ${}^7\text{Li}$ were present, but the production of heavier elements had to wait for the formation of the first stars ([27]).

⁶A notable exception is the temperature of neutrinos, which decoupled from baryons at $t \approx 1$ s [107].

⁷The time when the number density of electrons n_e dropped to one half is strangely called *recombination* ($t = 240\,000$ yr).

⁸This is simply because $\lambda_\gamma = 1/\sigma_T n_e$, with $\sigma_T = 8\pi/3(\alpha\hbar c/m_e c^2)^2 \approx 6.652 \times 10^{-29}$ m² being the Thomson scattering cross section.

tween 150 and 1000 Myr after the Big Bang); this event is labeled “First stars” in Fig. 2. Today we believe that the formation of structures proceeded from the smallest ones (stars and then galaxies) to the largest ones (clusters of galaxies) in a so-called bottom-up⁹ scenario, by amalgamation. This is only possible if *dark matter*, which constitutes 85 % of total mass of the Universe but does not interact with light and it is therefore transparent, is made by “cold” particles (i.e., traveling at speeds much smaller than c).

There is evidence that the very first objects that formed in the early Universe provided enough energy to ionize matter and produce free electrons, which were therefore able once more to interact with the relic photons through Thomson scattering: this event is usually called *reionization* (for a general introduction, see [68]). However, the density of the Universe at this time was far smaller than at the time of recombination, and therefore these new interactions¹⁰ were weaker: therefore, reionization has not been able to re-establish thermal equilibrium between matter and radiation anymore.

There is evidence of an acceleration in the expansion of the Universe [69], as shown by the asymptotic enlargement on the right labeled with “Dark energy?” in Fig. 2. This is the main experimental motivation for the presence of the cosmological constant Λ in Eq.2, whose value is presently estimated to be $1.19 \times 10^{-52} \text{ m}^{-2}$ [84]. We will not dig into this topic here.

Because of the double assumption of the presence of cold dark matter and dark energy, the standard model of cosmology is often called the Λ Cold Dark Matter model (Λ CDM).

2.3 How to look for evidence of the Big Bang model

In the previous paragraph, we have provided a brief description of the evolution of the Universe according to the SCM; it is now time to say something more about the way cosmologists have gathered the evidence necessary to build the model itself.

The finite speed of light gives a handy way to study how the Universe

⁹The opposite scenario (top-down) assumed that the first structures to be formed in the universe were the largest ones, and that smaller structures formed by disgregation of the formers.

¹⁰The mean free path scales as the inverse of numerical density, $\lambda \propto 1/n$, and that matter density obviously scales as $n \propto a^{-3}$ because of Eq. (1).

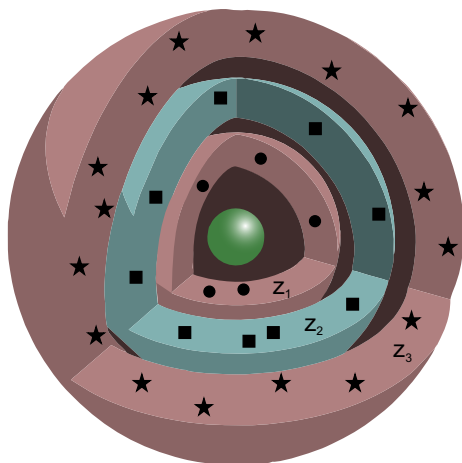


Figure 3: Subdivision of the space around Earth in classes of redshifts. The cosmological redshifts z increases with distance, but it is also a measure of the age of the objects. If we group the objects observed from Earth into classes of redshifts, each group will contain all the objects enclosed in a spherical shell of some radius, and the larger the radius, the older the objects. We show here the Earth as a globe in the middle of the figure, and a number of objects in the space around the Earth (circles, squares, stars), divided in three classes of redshifts $z_1 < z_2 < z_3$. Objects within the spherical shell z_1 (●) are younger than those in z_2 (■), while objects in z_3 (★) are the oldest and farthest. Therefore, measuring a physical quantity in each of the three classes permits to determine the dependence of that quantity on redshift, and thus on time.

looked in the past, as distant objects are observed today as they looked when the radiation we are receiving was emitted. Therefore, the farther the object we observe, the more ancient it is. Understanding the evolution of the Universe is in some sense the exercise of observing the space around us and categorize the objects within it according to their distance. However, since measuring redshifts (see Sect. 2.1) is much simpler than measuring distances, cosmologists group the objects according to their redshifts z by means of spectroscopic measurements, as the mathematical relation between z and distance (and age) is unique.

We refer now to Fig. 3, where we represent Earth as a sphere in the center of our observable Universe, and we categorize the objects we are

able to see from Earth according to their redshift. As we said before, since redshift measures distance, this categorization divides the space in spherical shells of some predetermined δz (the thickness of each spherical shell in Fig. 3). In the figure, we choose to consider three redshifts $z_1 < z_2 < z_3$, and we group the objects depending on the redshift bin they belong to (we use circles, squares, and stars for each group). Suppose that we want measure the density of matter ρ of the Universe, or some other physical parameter relevant for cosmology. We can estimate the value of ρ by analyzing the objects in each redshift bin, so that at the end we get three estimates: $\rho(z_1)$, $\rho(z_2)$, and $\rho(z_3)$. Since z can be used as a measure of time, this means that we can reconstruct the time dependence of ρ , or any other cosmological quantity, by observing objects on a broad range of distances!

The method of surveying objects at different redshifts is powerful, but it has its own share of problems. First of all, we need to be sure that the redshift we measure is purely due to the cosmological expansion of space due to Eq. (1), and not to some other physical effect. For instance, proper motion of celestial objects cause spectral shifts due to the Doppler effect. Typical speeds of stars and galaxies are in the range $10 \div 1000 \text{ km/s}$; this means that they can cause a blueshift/redshift $z \sim v/c \sim 10^{-5} \div 10^{-3}$ that is not cosmological. A way to solve this problem is to concentrate on very distant objects, so that the cosmological redshift is larger than 10^{-3} . Another problem is that observing a broad range of redshifts usually requires different technologies to be used in instruments. Consider for instance a survey that searches for celestial objects emitting¹¹ Ly- α lines, with $\lambda = 121.567 \text{ nm}$. In laboratory conditions, this line is in the UV range: to detect it, one would therefore expect to use UV detectors. However, because of cosmological redshifts, this line might well be in the optical or even infrared range, thus requiring a different kind of detector according to the redshift. Let's therefore consider the typical redshifts found in cosmological measurements, in order to understand the frequencies of the electromagnetic spectrum that are to be explored.

Apart from supernovae, and from rare notable exceptions (e.g., [53]), stars visible from the Earth show no redshift due to expansion ($z = 0$), and thus they can be used to study the present Universe. Superclusters of

¹¹The Lyman- α line is the $2 \rightarrow 1$ spectral line of Hydrogen, and it is commonly found in celestial bodies, due to the ubiquity of Hydrogen.

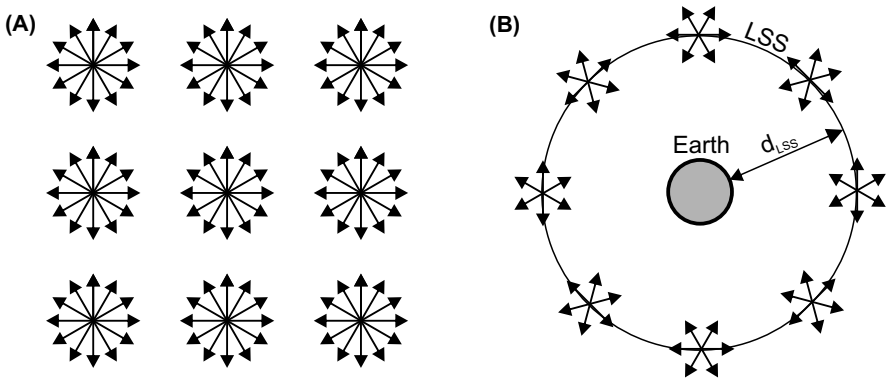


Figure 4: Physical origin of the Last Scattering Surface (LSS). **Panel A:** Under the hypothesis that decoupling was an instantaneous event, every point in space acted like a source of photons propagating freely in every direction. **Panel B:** Definition of the LSS. For an explanation, see the text.

galaxies, the largest gravitationally bound objects known today, are relatively close to us on a cosmological scale, but their redshift is observable ($z < 1.5$, see for instance [71]). Clusters of galaxies and quasars allow us to reach the age of reionization, as their redshifts can go beyond $z = 10$ (the most distant galaxy known so far, GN-z11, is at $z \approx 11.1$ [73]). However, what is most interesting in the context of this work is the existence of an “object” that has been produced only at a specific, very high redshift: the Cosmic Microwave Background (CMB for short), which is a radiation field at $z \approx 1100$. Because of its distance and age, the CMB is one of the most interesting subjects for cosmological studies: in the next section we present its origin and properties.

2.4 The Cosmic Microwave Background (CMB)

After the decoupling event, radiation was no longer in thermal equilibrium with matter and was essentially free to wander in the Universe with little interaction¹² since then. Therefore, it should be possible to observe

¹²Apart from the tiny interaction due to the reionization of the Universe, as we saw at the end of Sect. 2.2, there are a few other ways for the relic radiation to still interact with matter: for instance, through gravitational lensing and the Sunyaev-Zel’dovich effect. We are going to ignore these phenomena for the moment.

it, with a few caveats. First of all, the decoupling was an event happening *everywhere*, in the sense that every point in space can be considered as an emitter¹³ of freely-propagating photons at the time of decoupling, as shown in Panel A of Fig. 4. However, of all the photons that were left to propagate freely after decoupling, only those headed towards the Earth can be detected from an Earthly observer. Consider Panel B of Fig. 4: under the simplifying hypothesis of a non-expanding Universe, the primordial photons we can observe *today* are those that during the decoupling were at a distance $c(t_{\text{now}} - t_{\text{dec}})$, with t_{dec} the time of decoupling. This means that at any given time an Earthly observer can see only the photons emitted by a spherical shell, called the Last Scattering Surface (LSS). If the Universe is expanding, the formula for the radius of the LSS is more complex but the explanation is the same. Note that different observers define their own LSS: the LSS of an observer on the Earth is different from the LSS of an observer centered on some far distant Galaxy, as the position of the two shells in space is different.

The theoretical prediction of the presence of a relic radiation produced during the first stages of the evolution of the Universe was made in 1948 [4], but its detection happened only in 1965 with the observations made by Arno Penzias and Robert W. Wilson with the antenna at the Holmdel observatory in New Jersey ([76], see also Sect. 5.3). The relic photons are observed today as a quasi-isotropic radiation coming from the sky with a black-body spectrum at $T \approx 2.73\text{K}$ and a number density of photons $n_\gamma \approx 400\text{cm}^{-3}$. Recognizing its cosmological origin, the radiation itself has been called Cosmic Microwave Background (CMB). Although most of the observations of the CMB are carried within the Solar System, there is evidence that the CMB permeates the whole observable Universe [46,72]. The CMB is one of the most important pillars of the SCM, and its study permits to determine a wealth of parameters characterizing the primordial universe [62], as well as unexpected links with fundamental mathematical constants [40]. Many review papers are available on this subject [8,13,22,95,97].

To measure the properties of the CMB (or any scalar field) on the sphere, the most useful (and used) tool is the decomposition of its temperature field into Laplace's spherical harmonics $Y_\ell^m(\theta, \phi)$. The functions Y_ℓ^m

¹³Strictly speaking, this terminology is incorrect, as photons were not emitted but scattered. But the term is pedagogically more apt to help the reader in understanding the problem we are going to discuss in this section.

form an orthogonal basis of the Hilbert space of square-integrable functions on the sphere:

$$Y_\ell^m(\theta, \phi) = k e^{im\phi} P_\ell^m(\cos \theta), \quad (12)$$

where $-\ell \leq m \leq \ell$, k is a constant used to normalize the basis, and P_ℓ^m is an associated Legendre polynomial. This basis satisfies the orthonormality property:

$$\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} Y_\ell^m(\theta, \phi) (Y_{\ell'}^{m'}(\theta, \phi))^* \sin \theta d\theta d\phi = \delta_{mm'} \delta_{\ell\ell'}, \quad (13)$$

where δ_{ij} is the Kronecker's delta. Therefore, any square-integrable function $f(\theta, \phi)$ on the sphere can be decomposed into a discrete set of complex coefficients $a_{\ell m}$ using the series

$$f(\theta, \phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_\ell^m(\theta, \phi), \quad (14)$$

where the series converges in L^2 . The interesting fact of Laplace's spherical harmonics is that the angular scale of their oscillations is roughly¹⁴ $180^\circ / \ell$; therefore, the modulus of the complex number $a_{\ell m}$ quantifies the amplitude of perturbations in the function $f(\theta, \phi)$ at a specific angular scale.

The values of the $a_{\ell m}$ coefficients encode both the amplitude and the phase of the spherical harmonics on the sphere, but this information is somewhat redundant for cosmological purposes. Because of the assumption of isotropy, the phase is really not significant when we use the statistics of the CMB to derive the parameters of the primordial universe, as the phase determines the orientation with respect to some set of Cartesian axes, which is of course arbitrary. For this reason, cosmologists prefer to analyze the *power spectrum* C_ℓ instead of the raw $a_{\ell m}$ coefficients:

$$C_\ell = \langle a_{\ell m} a_{\ell m}^* \rangle, \quad (15)$$

where the $*$ symbol denotes complex conjugation, and the average operation $\langle \cdot \rangle$ requires some explanation. There are two ways to compute the average, which are applicable to quite different contexts:

¹⁴For $\ell = 0$, the $a_{\ell m}$ coefficients represents the average of f over the sphere, as $Y_{\ell m}$ is a constant in this case. This term is often called the *monopole*. If the field f represents the temperature on the sky of the CMB, then $a_{00} \approx 2.71$ K.

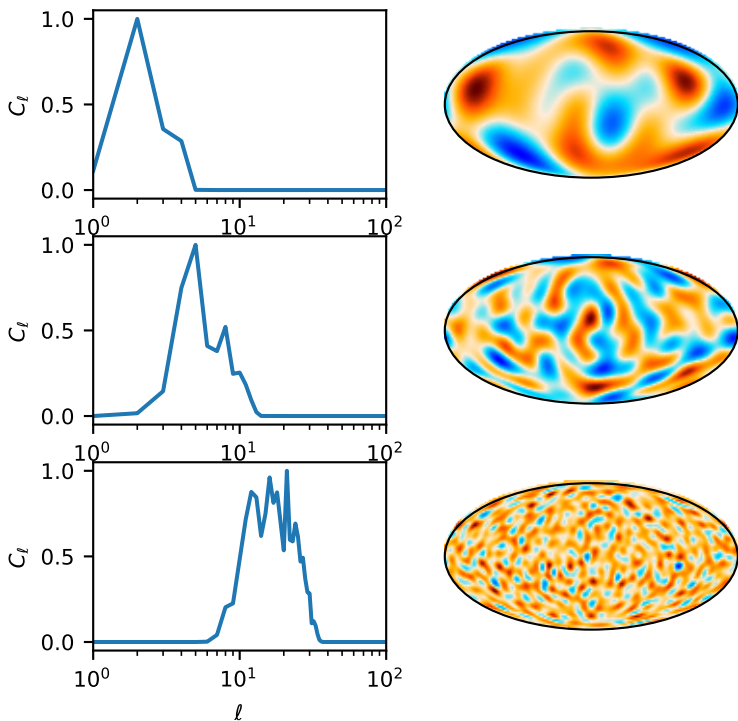


Figure 5: Relationship between angular features in maps and their power spectra. In the right column, three full-sky maps are shown using the Mollweide projection. In the left column, we show the power spectrum C_ℓ (in arbitrary units) of the map. The peak of the three spectra, from top to bottom, is at $\ell = 2, 5$, and 21 , which correspond roughly to angular scales of 90° , 35° , and 10° , as $\theta \sim 180^\circ/\ell$.

1. When one wants to estimate the power spectrum of the CMB from a model of the primordial Universe, one should ideally use this model to create a realistic picture of the CMB at the LSS around some observer P , and then make P vary over all the conceivable observers of the Universe. The average in Eq. (15) is meant to be calculated over all these observers. (Remember that each observer P has his own LSS, and therefore his own view of the CMB.)
2. On the other hand, when one wants to estimate the power spectrum of a map that has been measured by some instrument, the average has to be done over all the m values. As there are $2\ell + 1$ coefficients $a_{\ell m}$ for any fixed ℓ , this means that the average will be made on more and more numbers as ℓ increases; on the other hand, for small values of ℓ , the statistical error on the estimate of C_ℓ will be larger. This statistical error is called *cosmic variance*, and it is unavoidable, as we have yet to find a way to send probes to very distant galaxies and image the CMB from there! Mathematically, cosmic variance is expressed using the following formula:

$$\frac{\delta C_\ell}{C_\ell} = \sqrt{\frac{1}{f_{\text{sky}}(2\ell + 1)}}, \quad (16)$$

where f_{sky} is the fraction of the sky covered by the map; it is always smaller than 1, as the emission of the Milky Way at microwave frequencies makes impossible to estimate the emission of the CMB behind it.

Apart from cosmic variance, the characteristics of the instrument used to perform the measurement affect the error in the estimate of C_ℓ . The following formula provides an estimate for this [36]:

$$\delta C_\ell = \sqrt{\frac{1}{f_{\text{sky}}(2\ell + 1)}} \frac{\Omega \sigma_{\text{pix}}^2}{N_{\text{pix}} W_\ell^2}, \quad (17)$$

where Ω is the solid angle observed by the instrument, σ_{pix} is the RMS per pixel, N_{pix} is the number of pixels in the map, and W_ℓ is the so-called *window function*, which depends on the optical characteristics of the instrument; for an instrument observing the sky with a Gaussian beam,

$$W_\ell^2 = \exp\left(-\frac{\ell(\ell + 1)\theta_{\text{FWHM}}}{\sqrt{8 \ln 2}}\right), \quad (18)$$

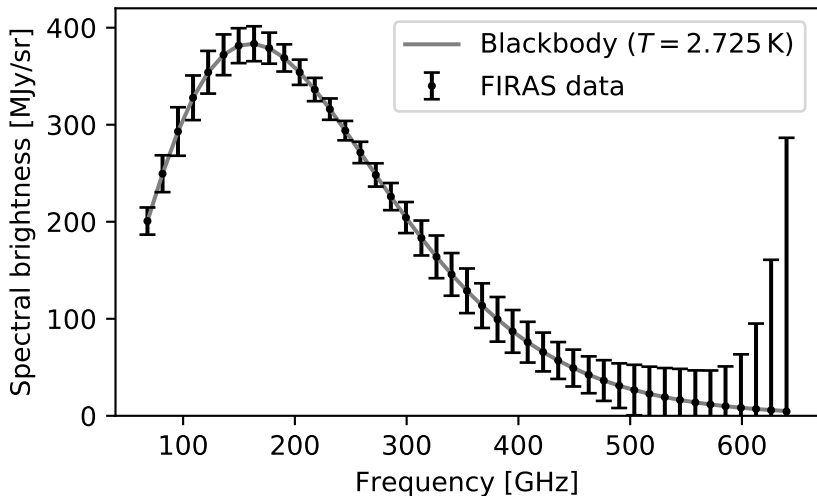


Figure 6: Spectrum of the CMB as measured by FIRAS. Error bars show the 1000σ error. Data have been taken from https://lambda.gsfc.nasa.gov/product/cobe/firas_monopole_get.cfm. The brightness is measured in MJy/sr, which corresponds to $10^{-20} \text{ W/m}^2/\text{Hz/sr}$.

where θ_{FWHM} is the FWHM of the beam (see Sect. 4.5 for its definition). In any realistic experiment, the error on the value of each C_ℓ is the sum of Eq. (16) and (17).

Now that we have introduced the basic tools used to quantitatively study the CMB, let's turn to the real measurement itself. The most important observables associated with the CMB radiation are the following:

1. The temperature associated with the blackbody spectrum and the presence of deviations from ideality [33]; the best measurement to date is the one produced by the FIRAS instrument onboard the COBE spacecraft [39], see Fig. 6.
2. The dependence of the blackbody temperature on the direction of observation (the so-called *temperature anisotropies*), expressed in terms of the C_ℓ power spectrum; the best measurement to date is the TT spectrum presented in [81], see Fig. 7.
3. The statistical distribution of blackbody temperatures associated with different directions of observation, with the primary purpose to hunt

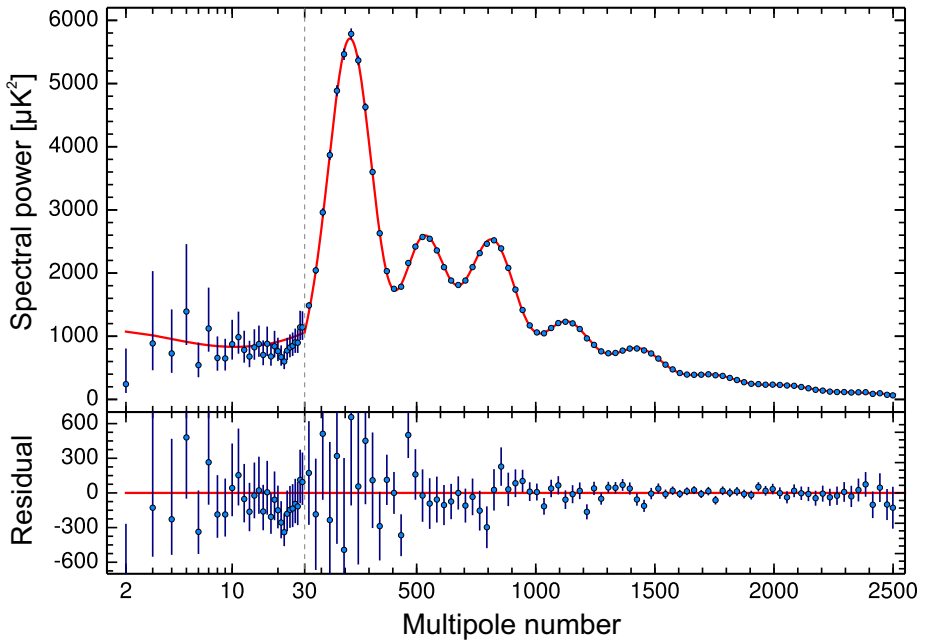


Figure 7: **Above:** Power spectrum of the CMB as measured by Planck [81] (blue dots) compared with the theoretical predictions of the Λ CDM model (red line). **Below:** The difference between the theoretical model and the measurements. In both plots, the quantity shown on the vertical axis is $\ell(\ell+1)C_\ell/2\pi$. For a very good explanation of the features of this plot, refer to [62]. For more information about Planck, see Sect. 5.3.

for deviations from a purely Gaussian distribution; so far, no evidence of non-Gaussianity have been found [86,93].

4. The dependence of the polarization angle and intensity on the direction of observation.

Each of these observables carries extremely interesting information. What we are interested in this work is the measurement of the last one, i.e., the detection of patterns in the angle of polarization of the CMB radiation measured on the sky sphere.

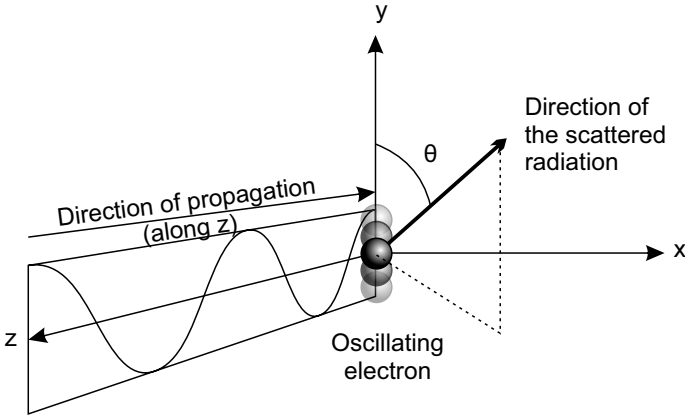


Figure 8: Depiction of Thomson scattering. A electromagnetic wave polarized along the $\hat{y}$ axis and coming from the $\hat{z}$ direction scatters on an electron e^- placed at the origin of axes. The electron oscillates up and down along the $\hat{y}$ axis, radiating waves that are polarized along $\hat{y}$ too. The intensity of the scattered radiation is proportional to $\sin^2 \theta$: it is maximum when $\theta = 90^\circ$, i.e., for directions perpendicular to the polarization vector (in this case, on the $\hat{x}\hat{z}$ plane).

3 Polarization of the CMB

To explain why we expect the CMB to be polarized, we must first recall the basics of Thomson scattering. This kind of scattering occurs when a photon interacts with a particle with charge q , whose rest energy mc^2 is much higher than the photon's energy $h\nu$ ($h\nu \ll mc^2$). What is interesting here is that the scattered radiation preserves the polarization direction of the incoming radiation, and that the scattered power per unit solid angle along direction $\hat{r}$ is

$$\frac{dP}{d\Omega}(\hat{r}) = \left(\frac{q^2}{4\pi\epsilon_0 mc^2} \right)^2 \frac{\epsilon_0 c E_0^2}{2} \sin^2 \theta, \quad (19)$$

where E_0 is the amplitude of the incoming wave and θ is the angle between the polarization direction $\vec{E}_0$ and $\hat{r}$. Note that there is no dependency on the frequency ν of the incoming wave, and that the power is maximum when the direction is perpendicular to the direction of the polarization vector (so that $\theta = 90^\circ$). Refer to Fig. 8 for the details.

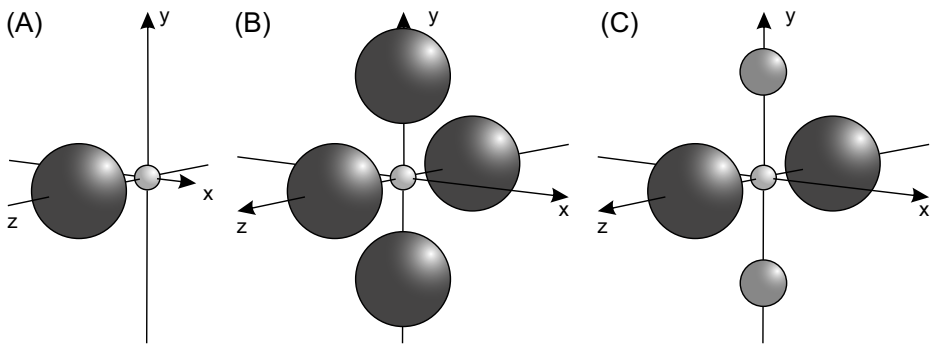


Figure 9: Following the same frame of reference as Fig. 8, we consider here what an observer on the x axis sees when a juxtaposition of waves interacts by Thomson scattering with an e^- placed at origin. **Panel A:** consider a source of unpolarized radiation along the z axis, represented by a sphere. The observer on the x axis is still able to measure radiation that is polarized on the y direction. **Panel B:** Consider four sources of unpolarized radiation, each with the same intensity, placed at $\pm x, \pm y$. In this case the contribution of each of the four sources cancels the polarization, so that the net radiation observed along x is unpolarized. **Panel C:** if the four sources of radiation radiate different powers, with the weakest being those along the y axis (represented by smaller, lighter spheres), then the observer will observe radiation that is polarized along the y direction.

So far, we have considered only a monochromatic plane wave; now we must turn to the most realistic case of a superposition of several plane waves, each with its own polarization and direction of propagation, as this is the situation with the CMB radiation. Intuitively, an uniform distribution of radiation around an electron does not scatter polarized radiation, as the polarization induced by each wave mode is averaged out. However, there is a peculiar spatial configuration, the *quadrupolar configuration*, which is able to produce polarized radiation: this happens when the radiation field around an electron has a variation on an angle of 90° . More rigorously, polarization arises if and only if the radiation field possesses a *quadrupolar* component. Figure 9 shows the basic idea; for a mathematical proof, see Sect. 2.2 in [8]. Mathematically, quadrupoles are defined mathematically as the Laplace's spherical harmonics Y_ℓ^m with $\ell = 2$. There are five possible configurations, corresponding to $m = 0, \pm 1, \pm 2$. In [45],

Hu & White provide a nice and intuitive explanation about the origin of each quadrupolar configuration in the Universe at decoupling. We shortly recapitulate the most important points:

- Adiabatic pressure oscillations (sound waves) produce quadrupoles with $\ell = 0$;
- Vorticity in velocity fields cause Doppler shifts, which in turn produce quadrupoles with $\ell = \pm 1$;
- Gravitational waves stretch the wavelengths of photons, producing quadrupoles with $\ell = \pm 2$.

All of these effects (pressure oscillations, velocity fields, gravitational waves) polarize the CMB radiation. The nice thing about these effects is the fact that the value of $|m|$ decouples them according to their physical origin.

It is important to underline the fact that polarization can only occur if the decoupling takes place in a short time, but not too short. If decoupling were an instantaneous process, there were no time for photons in a quadrupole around a point P to scatter in P to produce polarization. On the other hand, if decoupling were too slow, quadrupole anisotropies would smooth out while reaching thermal equilibrium, and thus polarization would not occur. Given the duration of decoupling, it can be estimated that roughly 10% of the CMB signal should be polarized [8], i.e., the temperature (brightness) of horizontally-polarized and vertically-polarized photons should be different by one tenth. This is an extremely difficult measurement to be made, because of the faintness of the signal; in the next section we will explain why it is worth to do it.

3.1 The Inflationary paradigm

The inflationary paradigm (or “Inflation”) is a theoretical framework able to explain the origin of the primordial seeds of all the structures we observe today, from the largest scales (CMB anisotropies to cluster of galaxies) to the smallest ones (stars and planets). The first inflationary theories were proposed in the early 1980s as a way to overcome a few problems in the traditional Big Bang scenario presented in Sect.2.2, namely:

1. Why is the CMB at LSS so uniform? Points at opposite directions of the LSS should have never been in causal contact, yet they show a temperature which differs by less than 10^{-5} ;

2. Why is the geometry of the Universe flat? If one studies the solutions allowed by the Friedmann equations, one readily discovers that the solution with $k = 0$ is *unstable*, i.e., any deviation from perfect flatness would become more and more dramatic as time passes.
3. How were structures formed in the first place? The standard Big Bang model is not able to explain the clumpiness of the observable Universe.
4. The Friedmann equations can perfectly model either an expanding or a contracting Universe. Why is our Universe expanding?

The inflationary paradigm solves all these problems by assuming that the Universe underwent a phase of accelerated expansion in the very first stages of the history of the Universe. This expansion enlarged microscopic quantum fluctuations to what will eventually become cosmological structures; plus, it put regions of space in thermal equilibrium out of casual contact. An exponential expansion would also have the effect of making the geometry flatter. For more information, see [9, 63]: Lineweaver’s review in particular is a bit dated but extremely clear to read.

Two features make the inflationary paradigm extremely interesting to study:

1. It is expected that a stochastic background of gravitational waves (GW) was generated in the first stages of the expansion. If inflation actually happened, the wavelength of these primordial GWs should have been expanded to cosmological sizes and might be large enough to be detected even today; if inflation is wrong, their wavelength would be way too small to be detectable. A detection of this GW background is considered to be a “smoking gun” of Inflation.
2. When Inflation took place, the average energy of particles in the primordial plasma was up to 10^{13} TeV [9], i.e., 12 orders of magnitude higher than the energy scales that can be tested in laboratories. Therefore, studying the Inflation might provide clues about a number of longstanding problems in fundamental physics, like the present irreconcilability between quantum mechanics and general relativity [65].

Proving that primordial GWs have been enlarged to scales that are detectable today would be a solid proof of the validity of the inflationary

paradigm. The wavelength expected for these gravitational waves is of the order of the Hubble radius, i.e., c/H_0 , which makes them too long to be detectable by GW interferometers like LIGO [61], VIRGO [106], or eLISA¹⁵ [5].

Thus, the most compelling evidence in favor of the inflationary paradigm would be the detection of relic GW. Fortunately, achieving this result is not impossible. Recall from our previous discussion (Sect. 3) that a tensor component like the one produced by primordial GWs produces polarization perturbations through quadrupoles with $m = \pm 2$. In Sect. 4.2 we shall see that this kind of perturbations has a characteristic signature that can be mathematically disentangled by scalar and vectorial perturbations ($m = 0, \pm 1$): this gives a tool to hunt for the relics of primordial GWs, thus potentially providing support to the inflationary paradigm.

4 Basic concepts of radio astronomy

Before we turn to the description of how the existence of relic GW can be assessed, we need to introduce a few concepts peculiar of radio astronomy. These will be useful to better understand how CMB experiments work.

4.1 Measuring the brightness of an object

For a very long time astronomers have been mostly interested in measuring the “brightness” of the sky, meant in a broad sense as a way of measure how much light (i.e., electromagnetic energy) is emitted by celestial objects. People have devised many ways to quantify the brightness of an object; widely used definitions in astronomy include the *magnitude* (logarithmic relative measure of the spectral brightness in a passband), the *spectral brightness* (energy power per unit surface per unit solid angle per frequency), the *spectral intensity* (energy power per unit solid angle per frequency), the *flux density* (energy power per squared meter), and so on.

CMB science usually quantifies the brightness of the CMB itself using the concept of *temperature*. This is due to the fact that the CMB spectrum is nearly a perfect blackbody (see Sect. 2.4), and it can therefore be fully parametrized by just one number, its thermodynamic temperature T . As

¹⁵LIGO and VIRGO can detect GW whose wavelength is in the range $10^3 \div 10^7$ m. For eLISA, the range is $10^8 \div 10^{13}$ m

microwave measurements of the sky are likely to catch radiation produced by non-blackbody sources as well, other definitions of temperature exists. An important definition is the *brightness temperature*, defined as follows. Given some measured spectral brightness I_ν (energy power per unit area per unit solid angle), the brightness temperature T_b associated with it is the temperature of a blackbody in thermal equilibrium with its environment¹⁶ that has the same spectral brightness as the observed object. Unfortunately, as it is often the case in astronomy, this basic and potentially unambiguous definition has been used at least in two incompatible ways:

- The most straightforward way to use the definition is to describe I_ν using Planck's blackbody law:

$$I_\nu = \frac{2h\nu^3}{c^2} \frac{1}{\exp\left(\frac{h\nu}{kT}\right) - 1}. \quad (20)$$

- However, Planck's law is difficult to invert analytically. Therefore, if $h\nu \ll kT$ (i.e., at low frequencies and/or high temperatures), Rayleigh-Jeans' law is often used:

$$I_\nu = \frac{2kT}{c^2} \nu^2. \quad (21)$$

When reading papers that quote microwave measurements using brightness temperatures, it is therefore important to understand which convention the authors used for T_b .

4.2 Measuring polarization

After we discussed in Sect. 4.1 how one can quantify the brightness of an electromagnetic wave, we turn here to the problem of measuring the characteristics of an incoming *polarized* wave. Consider a plane wave like the one in Fig. 8, which propagates along the $\hat{z}$ direction. Let's write its electric field $\vec{E}$ as

$$E_x = a_x \cos(\omega t - \phi_x), \quad E_y = a_y \sin(\omega t - \phi_y). \quad (22)$$

¹⁶Obviously, when observing a blackbody, its brightness temperature is the same as its thermodynamic temperature, assuming no attenuation in the propagation of light from the body to the observer.

(for the wave shown in Fig. 8, $a_x = 0$). We define the four *Stokes parameters* I , Q , U , and V as follows:

$$I = \langle a_x^2 \rangle + \langle b_x^2 \rangle, \quad (23)$$

$$Q = \langle a_x^2 \rangle - \langle b_x^2 \rangle, \quad (24)$$

$$U = \langle 2a_x a_y \cos(\phi_x - \phi_y) \rangle, \quad (25)$$

$$V = \langle 2a_x a_y \sin(\phi_x - \phi_y) \rangle, \quad (26)$$

where the average $\langle \cdot \rangle$ is meant to be calculated over a time span much larger than λ/c . The parameter I quantifies the average power per unit area of the wave (apart from a constant factor $\epsilon_0 c/2$, which we silently dropped), while Q and U characterize its linear polarization. Finally, V quantifies the amount of circular polarization, as it is nonzero when $\phi_x \neq \phi_y$. Because of the intrinsic symmetry of Thomson scattering, we do not expect the CMB to have circular polarization [70], so we are going to ignore the V parameter from now on. In fact, to properly measure the CMB polarization, it is enough to measure the Q and U parameters.

The mathematical properties of Q and U are peculiar, as they are not proper scalar quantities. In fact, it is easy to show that under a rotation by an angle θ they transform according to the equation

$$\begin{pmatrix} Q' \\ U' \end{pmatrix} = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{pmatrix} \begin{pmatrix} Q \\ U \end{pmatrix}. \quad (27)$$

Instead, the Stokes parameters Q and U are the components of the second rank, traceless tensor P :

$$P_{ab} = \begin{pmatrix} Q & U \\ U & -Q \end{pmatrix}. \quad (28)$$

For visualization purposes, one often defines a polarization pseudo-vector P with length $\sqrt{Q^2 + U^2}$ and angle $\arctan(U/Q)/2$, although this vector remains identical after a rotation by an angle π , and it therefore represents an orientation rather than a direction.

4.3 E/B polarization modes

All the formulas dealing with spherical harmonics in Sect. 2.4 are useful for decomposing *scalar* fields on a sphere: this is the case for the temperature of the CMB at the LSS. However, the approach can be generalized

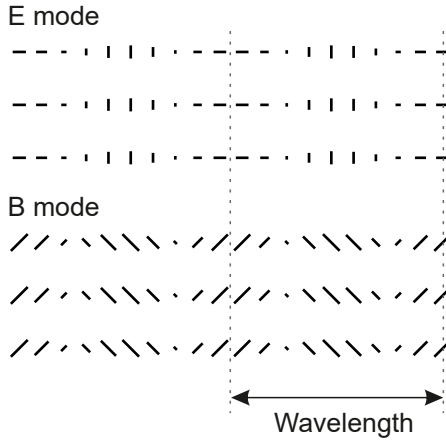


Figure 10: Depiction of pure E-mode (above) and B-mode (below) polarization on the plane. The former can be distinguished by the latter because of the lack of handedness: E-modes remain the same when reflected through a mirror, while B-modes flip. We show here just one Fourier mode, with some fixed wavelength along the left-right direction. Real-world polarization fields would be a superimposition of E and B modes, with different directions and wavelengths.

to vector and tensor fields on the sphere as well, so that the CMB polarization at the LSS can be decomposed into spherical harmonics as well. In the context of CMB polarization studies, [49] proposed the use of the symmetric trace-free tensor spherical harmonics $Y_{(\ell m)ab}^E$ and $Y_{(\ell m)ab}^B$ [28], so that the polarization tensor $P_{ab}(\theta, \phi)$ in Eq. (28) can be decomposed as

$$P_{ab}(\theta, \phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} (a_{\ell m}^E Y_{(\ell m)ab}^E + a_{\ell m}^B Y_{(\ell m)ab}^B). \quad (29)$$

Since E and B modes are defined in terms of Q and U using a non-local relationship, no detector can produce a direct measurement of E and B modes, regardless of the technology: the analysis of polarization data requires to measure Q and U and produce a sky map, and then to convert the map to E and B (typically in the harmonic space of the $a_{\ell m}$ coefficients).

The sets of $a_{\ell m}$ coefficients related to a temperature map are usually indicated with the subscript T . Therefore, if an instrument is able to measure I , Q , and U at the same time, then three sets of harmonic coefficients

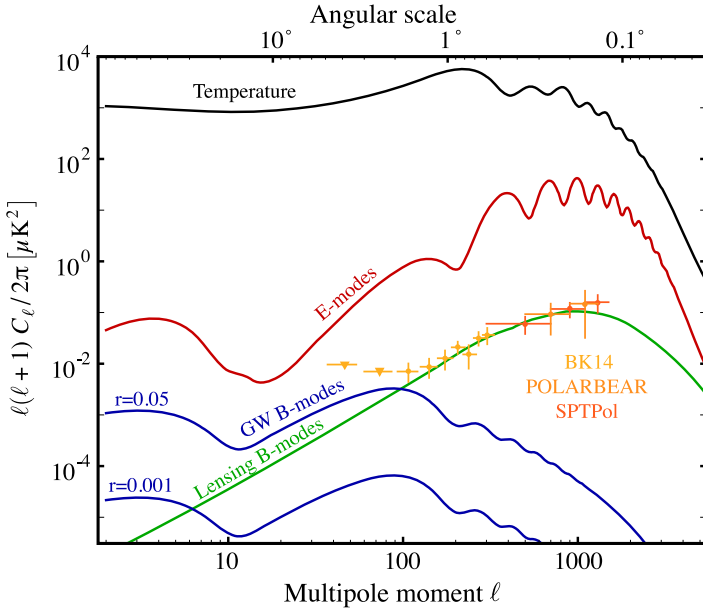


Figure 11: Theoretical predictions for temperature, E-mode, and B-mode polarization power spectra, taken from [1]. Note the significant difference in the absolute level of the three spectra, with the primordial B-mode spectrum being the faintest. Current measurements of the B-mode spectrum are shown for a few experiments: although the lensing spectrum has been measured, no detection of the primordial spectrum due to GW has been found so far.

can be defined: $a_{\ell m}^T$, $a_{\ell m}^E$, and $a_{\ell m}^B$. Thus, Eq. (15) can be generalized:

$$C_\ell^{\alpha\beta} = \langle a_{\ell m}^\alpha (a_{\ell m}^\beta)^* \rangle, \quad (30)$$

where α and β can be T , E , or B . Power spectra with $\alpha = \beta$ (i.e., C_ℓ^{TT} , C_ℓ^{EE} , C_ℓ^{BB}) are called *auto-spectra*, and they measure the power in temperature, E, or B modes for any ℓ . If $\alpha \neq \beta$ (e.g., C_ℓ^{TE}) are called *cross-spectra*, and they measure the amount of correlation between two modes: for instance, C_ℓ^{TE} might be large at some angular scale ℓ because there is a significant amount of power both in temperature and E-mode polarization. By symmetry, it is possible to prove that the primordial¹⁷ cross-spectra C_ℓ^{EB} and

¹⁷It is crucial to underline that this is true only for *primordial* polarization patterns. The

C_ℓ^{TB} must be consistent with zero at any angular scale ℓ [45]: this is often used as a test to assess the quality of experimental data.

We can now provide the clue about how to detect the existence of relic GW. Recall from the end of Sect. 2.2 that GW induce a polarized pattern because of the excitement of a quadrupole with $m = \pm 2$. It turns out that B-mode polarization can only be produced with this kind of quadrupole, as $m = 0, \pm 1$ can only produce an E-mode pattern. Therefore, to prove the existence of relic GW in the Universe during inflation, it is enough to measure a BB power spectrum $C_\ell^{BB} \neq 0$.

Unfortunately, measuring C_ℓ^{BB} is not a small feat, as Fig. 11 shows: the amount of power associated with B modes is significantly smaller than the power in E modes (which have been measured several times, see Fig. 21). What we have today is only a set of upper limits on the value of the BB spectrum, and these limits tell us that the BB spectrum is at least one order of magnitude fainter than the EE spectrum. This is usually quantified by the r parameter, which is the ratio between the power of primordial BB modes and the power of primordial EE modes in the CMB. Inflation theory does not give well defined predictions for r , but if inflation never happened we expect r to be zero; Fig. 11 shows two possible scenarios, for $r = 0.05$ and $r = 0.001$. The current upper bound for r is 0.12 at 95 % confidence [15].

4.4 Foreground emissions

One of the most difficult tasks in measuring the pattern of CMB polarization anisotropies is the proper disentanglement between polarization of cosmological origin and polarization due to objects between us and the LSS [35, 83]. The CMB community uses the term *foreground* to refer to any emission nearer than the LSS which is in the interval of frequencies 10–1000 GHz. As shown in Fig. 12, there is no frequency in this range where CMB polarization anisotropies are dominant¹⁸ over foregrounds. The immediate implication is that, in order to properly disentangle the

polarization pattern produced by Galactic emissions (see Sect. 4.4) can have nonzero C_ℓ^{EB} and C_ℓ^{TB} .

¹⁸This is not true for temperature anisotropies, where the CMB is dominant over other emissions when observed far from the Galactic plane. However, there are more unpolarized foreground emissions that make the measurement difficult, so much of what we say in this section is true for CMB temperature analysis as well.

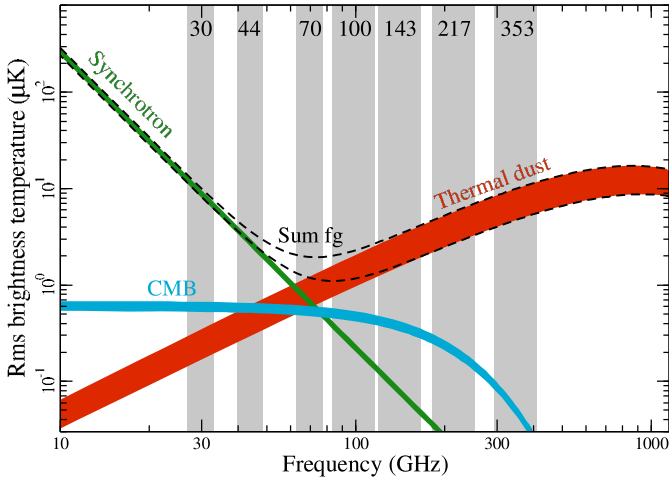


Figure 12: Polarized foregrounds measured by Planck, adapted from [83] (Fig. 51). For each foreground component, the full sky map has been smoothed to 40arcmin and the RMS of the pixels has been computed as a function of frequency, in order to measure the overall amount of anisotropies for $\ell \lesssim 250$. The intensity has been expressed as a brilliance temperature, using the Rayleigh-Jeans formula in Eq. (21). Vertical bars represent the bandwidth of the polarization-sensitive detectors.

CMB signal from foregrounds, one needs to have a multi-band measurement over the 10–1000 GHz range with exquisite sensitivity, in order to use the different spectral dependence of the foregrounds and of the CMB to disentangle them and obtain a clean, pure CMB map. The problem of separating the CMB from other emissions is called *component separation* [11,41,42,56,80,82,83,92], and it is a complex task that can be solved¹⁹ using several mathematical methods, exploiting the fact that the statistical properties and the spectral dependency of the CMB are quite different from the other emissions.

The foreground emissions that are expected to be problematic for CMB polarization measurements are the following (see also Fig. 12):

1. Cosmic rays trapped in the magnetic field lines of the Galaxy emit

¹⁹Outside the CMB world, this is often called the *Cocktail party problem*, where the task is to understand what a speaker is saying in a chatty room full of people. The words spoken by the speaker are the CMB signal, while the chatter represents the foreground emissions.

synchrotron radiation with a power law of the form $\nu^{-\beta}$ [18,23,24,34,96]; this radiation is intrinsically polarized.

2. Dust grains aligned by means of some mechanism (e.g., magnetic fields again) emit polarized radiation [6,24,105].

Lensing can be considered a foreground contamination, however it is of a fairly different nature than those described above. Foregrounds like thermal dust emission are caused by the presence of photons of non-cosmological origin in the electromagnetic radiation entering the optical system of an instrument. Lensing does not produce new photons, but it creates spurious polarization patterns when the photons are deflected by the gravitational wells of large scale structures during their journey from the LSS to the observer. In fact, lensing mixes power between E and B modes, so that a purely E-mode polarization pattern becomes an E/B pattern. The range of angular scales in the power spectrum that are affected by lensing is broad, but it peaks around $\ell \sim 1000$, thus at substantially smaller angular scales than those expected for inflationary B-modes. Lensing B modes have been detected in two ways:

1. It can be detected from a correlation analysis between the position of large scale structures on the sky sphere and the map of the CMB polarization [43,85,88,104];
2. Recently, the lensing BB auto-spectrum (Eq. 15) has been measured directly [14,52,87].

In order to detect primordial B-mode polarization, it is therefore necessary to clean the CMB polarization maps through a procedure called delensing [64,85].

4.5 Optical response of a detector

An ideal optical system would respond to radiation incoming from an infinitesimally small solid angle in the sky: in this way, the spatial information about the origin of the radiation would be fully preserved. However, interference and diffraction of electromagnetic waves make this situation impossible to achieve in practice. The situation is true for any field in astronomy, but it is particularly important for radio and microwave astronomy, because of the fact that wavelengths are often comparable to the

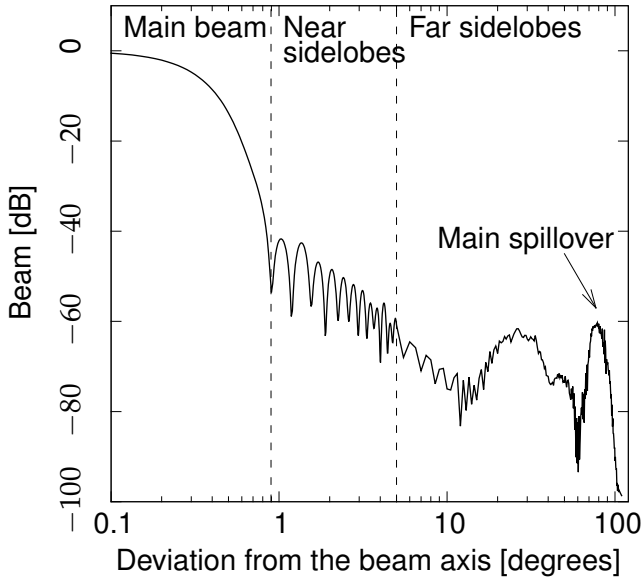


Figure 13: A typical beam response $\gamma(\theta, \phi)$ (Eq. 31), taken from [79]. Ideally, one wants the region of the main beam to be infinitely narrow and any other region as close as possible to zero. Note that the vertical axis is expressed in dB (logarithmic scale). See also Fig. 17 for a 3D representation of the response of a beam.

size of the optical elements of the instruments. In this section we are going to provide the basic terminology needed to understand the problem; Sect. 5.2 will discuss the most important technological solutions used for the kind of optical systems used in CMB experiments.

The optical response of an optical system is quantified by the so-called *beam function* $\gamma(\theta, \phi)$, which is used to express the temperature T measured by an instrument:

$$T(\theta, \phi) = \frac{\int_{4\pi} T_{\text{sky}}(\theta', \phi') \gamma(\theta', \phi') d\Omega'}{\int_{4\pi} \gamma(\theta', \phi') d\Omega'}. \quad (31)$$

The fraction represents a weighted average between the temperature of the sky and the beam response function $\gamma(\theta, \phi)$. Typically, $\gamma(\theta, \phi)$ is sharply peaked at the origin (the so-called *boresight direction*) and quickly goes to zero. The *pencil beam* idealization corresponds to γ being a Dirac's delta: this is an useful simplification when doing analytical calculations.

A typical profile for $\gamma(\theta, \phi)$ is shown in Fig. 13, which represents one of the beams of the Planck/LFI instrument. The angular width of the main beam at half its maximum height is called the Full Width Half Maximum (FWHM), and it is one of the most used parameters to characterize the resolution of an instrument. The value θ_{FWHM} is related to the maximum value of ℓ in the power spectrum C_ℓ (see eq. 15) that can be measured by an instrument, as any details of the CMB emission which is smaller than this size will be washed out by the weighted average operation in Eq. (31).

Polarization measurements require extreme purity in the optical response of the instruments. For instance, the beam should as symmetric as possible around the boresight direction, in order not to introduce spurious polarization. An important quantity that assesses the level of systematic error introduced by an optical element is its cross-polarization:

$$\text{XPD} = 10 \log_{10} \frac{X_\perp^2}{X_\parallel^2} = 20 \log_{10} \frac{X_\perp}{X_\parallel}, \quad (32)$$

where X is one component of the electromagnetic field (E or H) *exiting* the element, and $\parallel$ and $\perp$ refers to the orientation of the wave *entering* the element. Two more relevant quantities are the *return loss* RL and *insertion loss* IL:

$$\text{RL} = 10 \log_{10} \frac{P_{\text{refl}}}{P_{\text{in}}}, \quad \text{IL} = 10 \log_{10} \frac{P_{\text{out}}}{P_{\text{in}}}, \quad (33)$$

where P_{in} is the power entering the element, P_{refl} is the power reflected back by the element, P_{out} the power exiting the element. Generally speaking, the RL depends mostly on the geometrical design of the element, while IL depends mostly on the physical characteristics of the material used to build the element (e.g., the extent of Ohmic losses); in the ideal case $P_{\text{in}} = P_{\text{out}}$ and $P_{\text{refl}} = 0$, so $\text{RL} = -\infty$ and $\text{IL} = 0$.

4.6 Dealing with noise

The output of any detector will contain some statistical noise, which reduces the sensitivity of its measurements. In this section we explain how the stochastic noise of a radio receiver can be characterized and what is the impact on the quality of the measurement.

A first distinction to be made is between *correlated* and *uncorrelated* noise. The first kind refers to the fact that the fluctuations induced by the noise produce drifts that are significant at long timescales, and it is usually

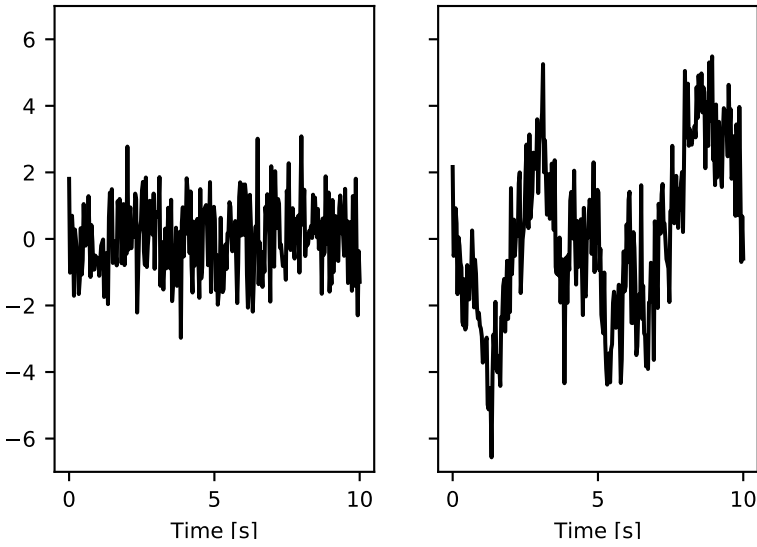


Figure 14: Example of a quantity measured by two devices with different noise properties. The quantity is supposed to be constant in time, and with zero average. **Left:** The output of the measurement done by a device whose noise is Gaussian (uncorrelated). The power is spread evenly among all the frequencies in the bandwidth. **Right:** The same measurement done by a device whose noise follows a $1/\nu^2$ spectrum (correlated noise, this is the case with $\alpha = 2$ in Eq. 34). The power is concentrated in the lowest frequencies of the band, as fluctuations on timescales of a few seconds have a larger amplitude than fluctuations faster than 1 s.

in the form of $1/f$ noise, where the power spectral density (PSD, power per frequency interval) associated with the noise follows the law²⁰

$$P(\nu) = P_0 \left(\frac{\nu_0}{\nu} \right)^\alpha, \quad (34)$$

with P_0 the PSD associated with the noise in the band $[\nu_0, \nu_0 + d\nu]$, and α a coefficient. For microwave detectors, the value of α is typically in the range 1–2. Uncorrelated noise has a simpler power law, because the power

²⁰The name “ $1/f$ noise” refers to the fact that in many textbooks the frequency ν is indicated as f . We prefer to keep using ν here.

is constant (independent of the frequency):

$$P(\nu) = P_0. \quad (35)$$

Measurements of the power associated with an electromagnetic field are always characterized by the presence of uncorrelated noise. If the frequency is high enough to consider the field to be made of discrete particles, then the noise follows a Poisson distribution; otherwise, the noise is Gaussian.

The sensitivity of an array of N receivers is given by the following formula [108]:

$$\delta T = \frac{T_{\text{sys}}}{\sqrt{N}} \sqrt{\frac{1}{\Delta\nu \tau} + \frac{\delta G}{G}}, \quad (36)$$

which provides the error in the estimate of the temperature T for a given pixel in the sky. In this equation, T_{sys} is the total input of each receiver (the sum of the signal entering the instrument plus any noise), $\Delta\nu$ is the effective bandwidth, τ is the integration time, and $\delta G/G$ quantifies the stability of the amplifiers used in the system. It is clear that there are several ways to improve the sensitivity of an instrument:

1. Increase the number of detectors. Missions like Planck (Sect. 5.3) implemented $\sim 10^2$ detectors, while future efforts like LiteBIRD will implement 10^5 detectors.
2. Increase the bandwidth $\Delta\nu$. Typical bands in CMB experiments are of the order of $\Delta\nu/\nu \sim 20 \div 30\%$.
3. Increase the integration time. This is especially easy for ground-based and space missions, where the overall time can be a few years (to be divided by the number of pixels in the sky map), but it can be a serious limit for balloon-borne experiments (Sect. 5.3).
4. Make the instrument as stable as possible, in order to minimize $\delta G/G$. Since gain fluctuations are typically due to thermal instabilities, this means that the instruments must be kept cooled using stable cryocoolers. The level of $\delta G/G$ is typically of the order of 10^{-2} – 10^{-3} .

In the next section we will see how real experiments implemented the techniques sketched here.

5 The state-of-the-art in CMB polarization measurements

Now that we have introduced the necessary terminology, we can turn on the discussion of the present state of technologies used to implement CMB experiments.

5.1 Detector technologies

To measure the flux of electromagnetic energy in the microwave region of the spectrum, a few technologies are available. Traditionally, CMB experiments have used either High Electron Mobility Transistors (HEMTs) or bolometers, although new devices are being built and tested. In this section we provide an overview of each of them, listing their advantages and shortcomings. Refer also to Fig. 15, which groups the experiments according to their technology.

5.1.1 HEMTs

The flux of radiation can be measured using radiometric receivers based on HEMT amplifiers. In this case, the radiation is first amplified and filtered by low-noise devices, and it is then converted into a voltage and measured by an Analog-to-Digital (ADC) electronic component. The advantage of this kind of measurement is the fact that the device can measure both phase and amplitude of the wave (*coherent detector*).

The most critical part of a radiometric receiver is the low-noise amplifier. The most widely used amplifier for CMB applications is the High Electron Mobility Transistor (HEMT). Being able to measure the phase permits to use HEMTs for direct measurements of Q and U (see Eq. 24 and 25); however, it has the cost that quantum noise puts a lower bound on the minimum achievable noise of the detector. The system noise temperature of a detector working at frequency ν is

$$T_{\text{sys}} = \frac{h\nu}{k \log 2}, \quad \Rightarrow \quad \left(\frac{T_{\text{sys}}}{1 \text{ K}} \right) = \left(\frac{\nu}{20 \text{ GHz}} \right). \quad (37)$$

This implies that HEMTs work better at low frequencies. For CMB measurements, this limit is typically well below 200 GHz: at higher frequencies, other technologies can be substantially more performant [59]. You can clearly see this in Fig. 15.

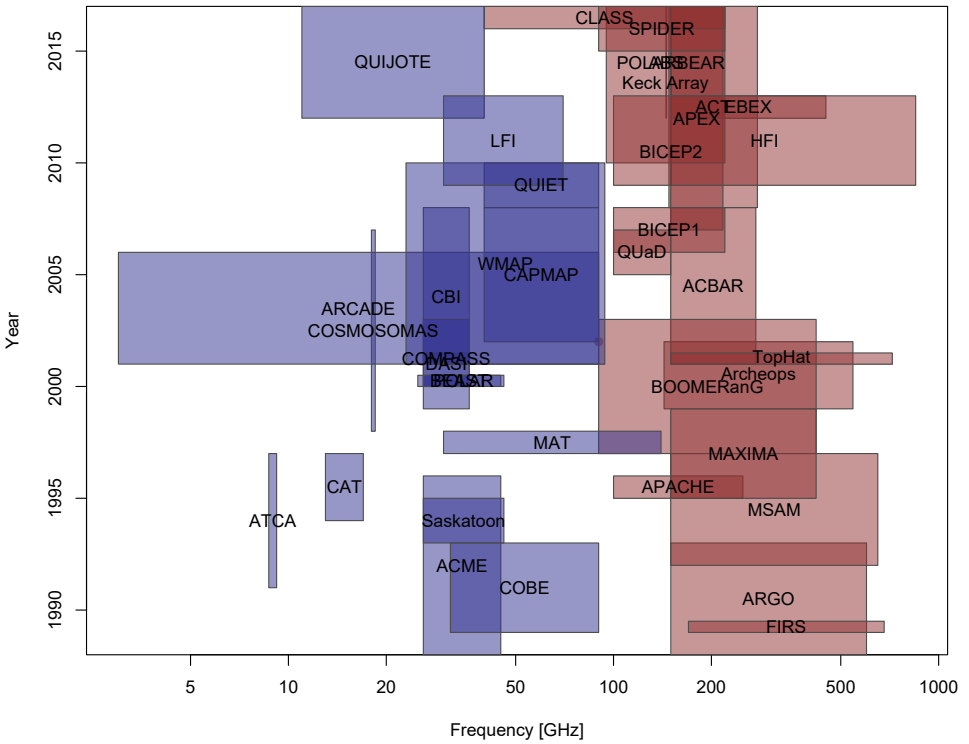


Figure 15: Overview of the most significant CMB experiments done in the last decades. For each experiment, the years of activity are shown along the vertical axis, while the frequency coverage spans the horizontal axis. The color identifies the technology used for the detectors.

5.1.2 Bolometers

A bolometer is a detector made by the combination of an absorptive element, a thermal reservoir and a thermometer, as shown in Fig. 16. The thermometer is placed on the absorptive element, such that when the element absorbs electromagnetic power, the thermometer (which is usually either a traditional thermometer or a Transition-Edge Sensor, TES) can measure the temperature increase. The role of the cold reservoir is to dissipate the excess heat absorbed by the element. Mathematically, the relationship between the temperature variation ΔT and the power P absorbed

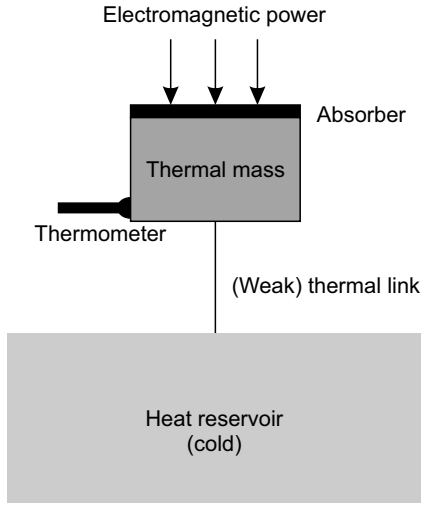


Figure 16: How a bolometer works. Incoming radiation is absorbed by a thermal mass, which increases its temperature by an amount ΔT . This temperature variation is measured by a thermometer and can be used to derive the absorbed power. A thermal mass is used as a sink for excess heat; in order to work, it must be kept at low temperatures. The thermal link between the mass and the reservoir must be weak, o

by the bolometer is the following:

$$C \frac{dT}{dt} + G\Delta T = P, \quad (38)$$

where C is the specific heat of the thermal mass, G is the thermal conductivity between the mass and the reservoir, and P is the absorbed power. If we imagine that the power P is constant in time, after some time equilibrium will be reached, so that

$$\Delta T = \frac{P}{G}. \quad (39)$$

Therefore, once ΔT is measured, one can readily estimate P . If the flux P goes abruptly to zero, we can estimate the time needed to reach the equilibrium again:

$$\Delta T(t) = \frac{P}{G} e^{-t/\tau}, \quad (40)$$

where $\tau = C/G$ is called the *time constant* of the bolometer. The value τ is related to the time scale needed by the bolometer to reach equilibrium, and it is an intrinsic property (and limitation) of each bolometric detector. In order for the bolometer to be fast (i.e., to output as many measurements as possible in a fixed time), one needs small values for τ , which typically means large conductivities. However, if G is large then ΔT in Eq. (39) is small and therefore difficult to detect: designing a good bolometer means to find the best trade-off between high data rates and sensitivity.

The fact that the energy of each photon $h\nu$ is converted into thermal energy implies that any information about the phase of the incoming wave (ϕ_x and ϕ_y in Eq. 22) is lost. This is both a blessing and a curse. It is an advantage because this means that there is no quantum lower limit on the noise achievable by a bolometer, but it means that a single bolometer cannot be able to directly measure the Stokes parameters Q and U . To measure the polarization of radiation, one must therefore put some polarization filter in front of the bolometer. Two typical choices are:

1. Using more than one bolometer associated with polarized filters provides enough information to estimate I , Q , and U at the same time. The most common configuration is to use 4 bolometers, each paired with a polarized filter, and to align the filters at angles 0° (A), 45° (B), 90° (C), and 135° (D). It is easy to prove that the pair A-C measures I and Q , while the pair B-D measures I and U [48];
2. Half-wave plates (HWP) are polarization filters put in continuous or stepped rotation, so that the time stream of measurements of one bolometer can be used to quantify both Q and U [58, 103]. This technique permits a better control of systematics, like $1/f$ noise, but it is technologically more demanding.

5.1.3 Kinetic Inductance Devices

A Kinetic Inductance Device (KID) is a novel kind of detector that measures the flux of radiation falling over a strip of superconducting materials by detecting the breaking of Cooper pairs [29]. The detector employs the fact that the kinetic inductance of a superconducting strip is inversely proportional to the density of Cooper pairs: a KID implement a microwave resonator by pairing the strip with a capacitor and measuring the variation in the resonant frequency. The two major potential advantages of

KID detectors over HEMTs and bolometers are the fact that they can be easily multiplexed²¹ in order to form large arrays of detectors, and that their response is usually faster than bolometric detectors (the τ constant in Eq. 40) by one order of magnitude and more.

KID detectors are used in the IRAM KIDs array (NIKA2), an instrument installed in October 2015 at the focus of IRAM’s 30 m telescope at Pico Veleta (Spain). This instrument implements 1000 detectors at 150 GHz and 4000 detectors at 240 GHz [20,21]. The GroundBIRD experiment [102] is expected to make use of this technology as well [74].

5.2 Optical systems for microwave measurements

After the general introduction about the optical response of an instrument, in Sect. 4.5, we now turn to a more precise definition of the problem. In this section we will provide a number of solutions that can be employed in order to reduce the systematic effects caused by the optics, as well as to achieve the desired angular resolution and polarization purity.

An optical system is a device whose purpose is to collect the light coming from the sky and bring it to the detectors. As we saw in the discussion regarding Eq. (31), it is characterized by its radiation pattern $\gamma(\theta, \phi)$, which is a three-dimensional quantity involving the variation of power as a function of the spherical coordinates θ and ϕ . The pattern has a main lobe (maximum radiation) in the boresight direction ($\theta = 0$) with sidelobes in other directions. Any field pattern can be presented in three-dimensional spherical coordinates, as in Fig. 17, or by plane cuts through the main-lobe axis, as we saw in Fig. 13.

Following the light from the sky toward the detector, CMB optical hardware typically includes some or all of the following elements: reflectors, lenses, band defining filters, amplifiers, detectors, and feedhorns or antennas. The order of some of these elements may vary somewhat. Intermediate elements, such as a vacuum window are generally not considered part of the optical system although they do have to be considered as an optical element during the design of the system.

Due to the wave nature of light, no image produced by an optical sys-

²¹Multiplexing refers to a technique where the output of several detectors (e.g., a voltage) is assembled into one signal travelling along the same line. This permits to perform measurements using only a few output cables, thus simplifying the design of the electronics.

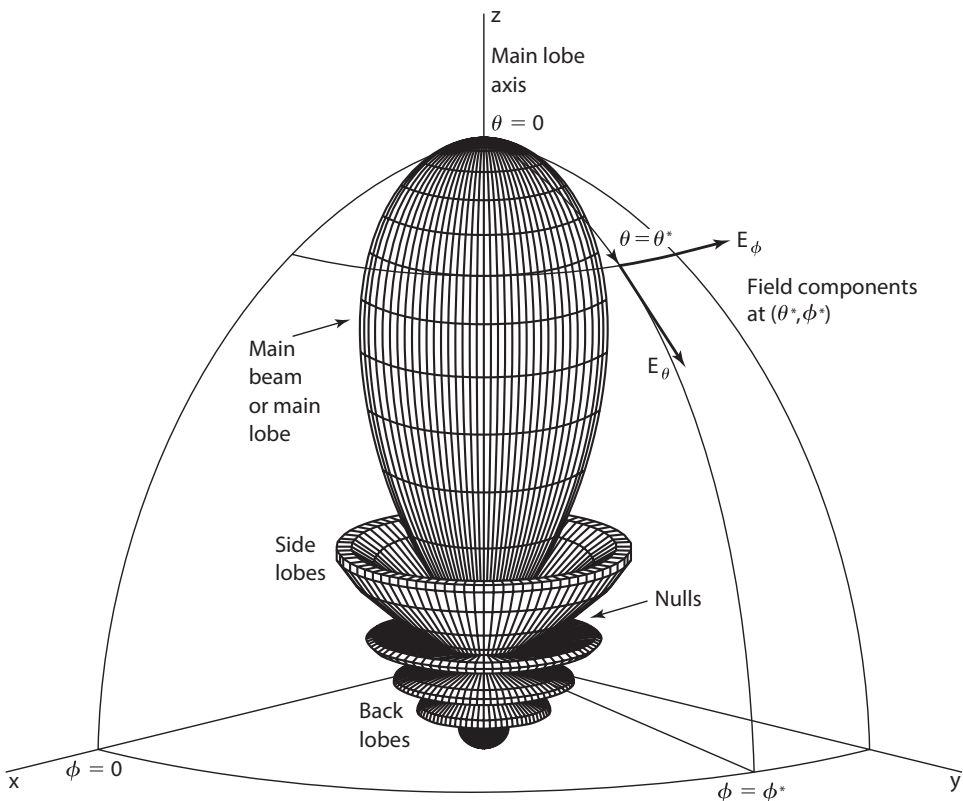


Figure 17: Three-dimensional pattern of a directional antenna with maximum radiation in the z -direction at $\theta = 0^\circ$ (line of sight). Most of the radiation is contained in the main beam.

tem is perfect in the sense that all light is brought to a single focus point of infinitesimal size. Since an optical system suffers from diffraction, the light will actually form a small disk surrounded by fainter rings, called diffraction rings. This is known as the Airy disk. The diffraction is caused by interference of light at the aperture of the telescope. In a perfect telescope 84% of the light goes to the central Airy disk and 16% to the rings, and it is impossible for more light to go into the central disk. In Sect. 5.2 we introduced the FWHM, which provides a measurement of the *angular resolution* of the telescope. Due to diffraction, the angular resolution limit for an optical device is

$$\theta \simeq 1.22 \frac{\lambda}{D}, \quad (41)$$

which is valid for a circular aperture uniformly illuminated.

The resolution of a CMB telescope is easiest to think about in the time reversed sense. If we imagine that a detector element emits radiation, then the optical elements in the receiver direct that beam towards the sky or onto the primary reflector. The aperture of the primary optical element of the telescope (e.g., a feedhorn that launches a beam to the sky, a lens, or a reflector), measured in units of wavelengths, determines the resolution of the instrument.

The connection between the spatial size of the beam at the primary optic and the resolution can be understood through the Fraunhofer's diffraction relation [17]:

$$\psi(\theta) \propto \int_{\text{aperture}} \psi_a(r) e^{kr \sin \theta \cos \phi} r dr d\phi, \quad (42)$$

where $\psi_a(r)$ is the scalar electric field (i.e., one component of the electric field) in the aperture or on the primary optic, and $\psi(\theta)$ is the angular distribution of the scalar electric field in the far field ($d \gg 2D^2/\lambda$). The integral is over the primary reflector or, more generally, the aperture.

The normalized beam profile is given by $\gamma(\theta) = |\psi(\theta)|^2 / |\psi(0)|^2$. That is, if the telescope scanned over a point source very far away, the output of the detector as measured in power would have this profile as a function of the scan angle θ . In angular dimensions, the beam profile is most often characterized by a full width at half maximum, or twice the angle at which $\gamma(\theta) = 1/2$.

One of the key characteristics for any optical system is the "throughput" (or "A-Omega") of the system. It is a measure of the total amount of radiation that can be collected by an optical system. This quantity is conserved for any lossless optical system, and it has a specific value for a system that supports only a "single mode" of propagating radiation²²

$$A\Omega = \lambda^2, \quad (43)$$

where λ is the wavelength of the radiation and A is the effective area of the aperture. The relation gives a handy conversion between the system's

²²A mode is a particular spatial pattern of the electromagnetic field. Experimentally, the selection for operating in a single mode is typically achieved by having a waveguide somewhere along the light path, typically at the entrance to the detecting element. The waveguide is essentially a high-pass filter, selecting the lowest frequency that can pass through the system.

effective aperture, the angular extent of the beam and the frequency of interest for single mode optical systems.

Other key parameters are the focal length and the f -number. For a given optical system, the *focal length* is the distance over which initially parallel rays are brought to a focus. We can define also the *f-number*, $F\#$ (or focal ratio), which is the ratio of the focal length f to the diameter of the aperture D ,

$$F\# = \frac{f}{D}. \quad (44)$$

It is a dimensionless number which gives a quantitative measure of the system speed²³. Note that rapid telescopes have small f -numbers. A smaller focal ratio means a wider field of view, hence it allows more light to reach the focal plane. At microwaves, small focal ratio means small feedhorn aperture, resulting in more compact focal plane units. Then, the more a telescope is rapid the more the feedhorn may be small in diameter. Moreover, a telescope with a small focal ratio is typically shorter and lighter than longer focal ratio telescopes.

For a given aperture size, i.e for telescopes with the same resolution, the performance of the optical system is measured in terms of the ‘diffraction limited field of view’ (DLFOV), which is the portion of the focal surface across which the optical performance is diffraction limited. The Strehl ratio is commonly used to determine whether the optical performance is diffraction limited: a system that provides a Strehl ratio larger than 0.8 at a particular field point is generally considered diffraction limited at that field point.

There are at present multiple proposed configurations for an instrument to measure the B-mode polarization of the CMB. Schemes range from bare corrugated horns aimed directly at the sky to large-aperture reflector telescopes, and include interferometers and smaller reflector/refractor systems. Each optical configuration is optimized for a specific scientific purpose and the choice is intimately connected with the desired noise performance. Once a resolution is chosen, one must consider whether to use a reflective or refractive system, a combination, or perhaps only a feedhorn.

In the simplest feedhorn-only system, the far-field beam shape, its angular size, and the band-width are determined by the shape of the feed-

²³The f -number is sometimes spoken as the speed of the optical system since the photographic exposure time is proportional to the square of the f -number.

horn and the waveguide components attached to it. However, optical system that use only feedhorns are limited to a coarse resolution. A horn with a reasonably large effective aperture diameter of 15 cm produces a beam size of $\sim 6^\circ$ and $\sim 1^\circ$ at 30 and 150 GHz, respectively. More complete studies show that it is difficult to produce a beam with θ at 150 GHz with a corrugated structure of reasonable volume [7].

Nowadays, CMB experiments use telescopes in order to increase the angular resolution. In the broadest terms, there are two classes of telescopes: refracting and reflecting. Both are usually associated with filtering and shielding structures.

5.2.1 Reflector-based optical systems

If we consider reflector-based systems there are several design choices, which are commonly divided into two groups:

Centered (or “on-axis”) optical system, where the reflectors are made from central portions of the typically conic sections of revolution that describe the surface shape. As a consequence, such systems naturally have central obscurations.

Decentered (sometimes also called “off-axis”, or “offset”) optical system, where off-axis portions of the conic sections are used, the reflectors are not centered on each other’s axis of symmetry, and there is no self obscuration by the reflectors.

For a fixed entrance aperture diameter, centered systems are more compact compared to decentered systems. However, they have lower aperture efficiency and are more prone to scattering of radiation to side-lobes caused by either diffraction from the edges of the central obscuration or by beam-intercepting supports of the secondary reflector. Decentered systems that have an intermediate focal point are typically easier to baffle compared to centered systems, and are thus less prone to stray radiation. As a consequence, virtually all CMB telescopes use decentered systems.

With two reflector systems, the classical configurations are either a Cassegrain (parabolic primary and hyperbolic secondary) or a Gregorian (parabolic primary and elliptical secondary). The primary source of aberrations in both Gregorian and Cassegrain systems are astigmatism and coma. Therefore a good design starting point are those designs that provide some cancellation of either of these aberrations, at least at the center

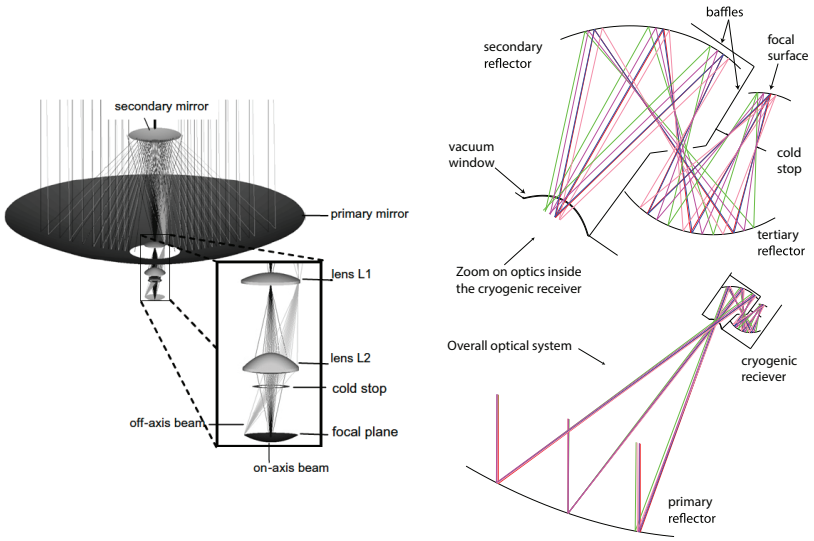


Figure 18: Examples of centred (left) and decentered (right) CMB optical systems. The centred, two-reflector Cassegrain system is that of the ground-based QUAD instrument [75]. The decentered, 3m reflector Gregorian system is that of the MAXIMA [91] balloon-borne experiment.

of the FOV. In an aplanatic Gregorian, making the primary slightly elliptical and the secondary a more eccentric ellipsoid than in the classical Gregorian cancels coma. The Planck optical system is based on an aplanatic Gregorian design.

Another innovative two reflector system that provides a particularly large FOV is described by [37]. This system, which has become known as the “Crossed Dragone” configuration, began to be adopted by some CMB experimenters in the mid-2000. The crossed-Dragone system has a parabolic primary and hyperbolic secondary and cancels astigmatism, coma, and spherical aberrations at the center of the field of view. The first CMB experiment that has made measurements with a crossed-Dragone system was QUIET [47] (see Fig. 19).

5.2.2 Refractor-based optical systems

In refractive systems, the absence of a central obscuration, combined with the on-axis nature of the optics provides large DLFOV. BICEP was the first

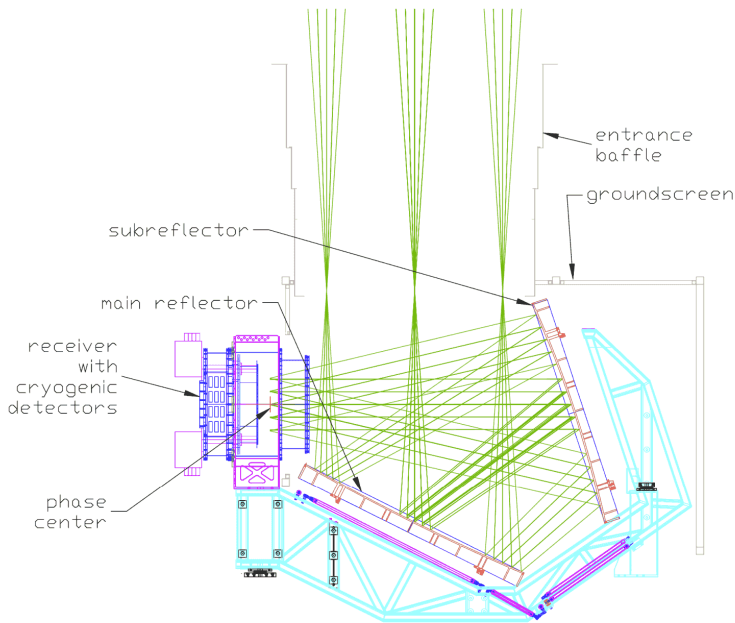


Figure 19: The crossed-Dragone telescope of the QUIET experiment. The primary reflector is 1.4 m in diameter.

CMB experiment to implement a refractive only system (see Fig. 20).

Usually the lenses used in these systems can be divided into high-index lenses, made of silicon ($n \simeq 3.4$) or alumina ($n \simeq 3$), and low index lenses, made of polyethylene ($n \sim 1.5$). High-index lenses are characterized by a high thermal conductivity that facilitates cooling and lowers the operating temperature of the optics, which in turn lowers the loss and thermal emission of the optics. On the other hand, polyethylene has been successfully used as a lens material in several CMB experiments. It is manufactured in multiple grades: low density (LDPE), high density (HDPE), and ultra high molecular weight (UHMWPE). These have similar physical and optical properties. This material makes AR coating easy and its good machinability makes large plates easily available; however, the low thermal conductivity of polyethylene makes the cryogenic design challenging for the larger diameter optics.

Practical considerations limit the use of a refractive-only system. One is the weight of the lens. The density of ultra-high molecular weight polyethylene is 0.95 g/cm^3 , making a 1 m diameter lens weigh around

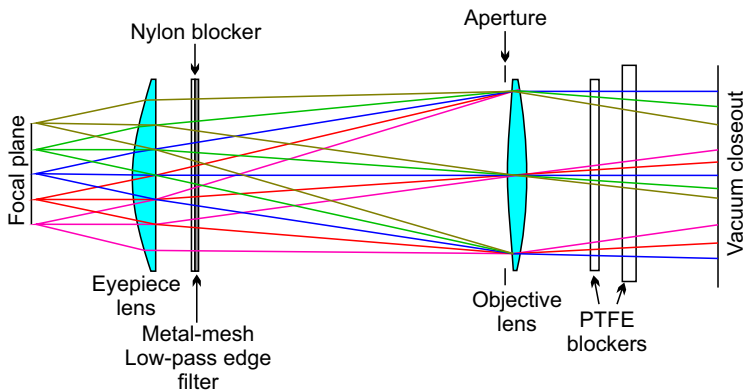


Figure 20: The all-refractive optical system of BICEP [51, 109]. The two lenses and other filters are all maintained at a temperature of 4 K.

100 kg. In contrast, the 1.4 m diameter WMAP primary reflector weighed only 5 kg.

5.2.3 Shielding structures

A common theme for all CMB telescopes is controlling and understanding the sidelobes. Invariably antenna patterns have non-vanishing response beyond the design radius of the optical elements. Unwanted power that is not originated by sources located in the main beam direction, named *straylight*, enters detectors through the sidelobes and it generates a signal that may be undistinguishable from CMB fluctuations in the main beam, as the telescope scans the sky. A particularly important source of straylight is the ground, as Earth has a temperature of ~ 300 K. To control the sidelobes the instrument uses shields and baffles to absorb and redirect spillover radiation.

Three techniques have been used most widely to control sidelobes, sometimes in combination with each other: (i) use of specifically-shaped feedhorns, (ii) use of an aperture stop to control and define the illumination on the primary reflector, (iii) use of shields and baffles to absorb and redirect spill-over radiation.

There are two basic types of baffles: absorptive and reflective. Absorptive structures (i.e. eccosorb lined aluminum) absorb straylight, but emit as a blackbody at the temperature of the baffle. Reflective structures (i.e. aluminum) redirect sidelobes to the sky, leading to a lower loading, but

reducing the understanding of the optical response.

5.2.4 Filtering structures

CMB detectors are subject not only to in-band photon loading, but also to radiative IR loading from warm optical elements such as the window, lenses, and telescope and receiver structures. In-band IR loading increases noise, whereas out-of-band IR loading makes it hard to keep temperatures cold and stable. This problem is particularly severe for broad-band detectors. Therefore, various reflective and absorptive optical filter technologies are employed to mitigate this infrared radiation.

Metal-mesh filters are reflective filters made embedding metal mesh on a dielectric. They are commonly low pass filters.

Nylon and Teflon filters are absorptive filters. These materials have strong absorption in the THz and IR region, yet acceptable transparency to mm-waves, making them useful for absorptive IR filtering. Teflon has a higher cut-off frequency and higher mm-wave transparency and thus has often been used as a first stage filter. Nylon has a lower cut-off frequency but also higher in-band losses, so it is generally used on colder stages. The problem is that big filters warm up in their middle region, causing a loading.

Alumina filters are attractive for CMB experiments due to the high mm-wave transmission, high IR absorption, and high thermal conductivity of alumina at cryogenic temperatures. In particular, the high conductivity prevents the center of the filter from heating when loaded with incident radiation. The optical and thermal properties, as well as the commercial availability of large diameter plates make alumina a promising IR blocking filter candidate for future CMB experiments.

Silicon substrate filters are hybrid filters based on reflective frequency-selective structures patterned on silicon substrate.

Foam based infrared blocking filters, such as stacks of Zotefoams and radio multi-layer insulation (RT-MLI)-based filters are constructed using commercially-available thermally insulating foam. Heat sinking to a cryogenic stage is not required for this filters to work effectively, simplifying the installation process.

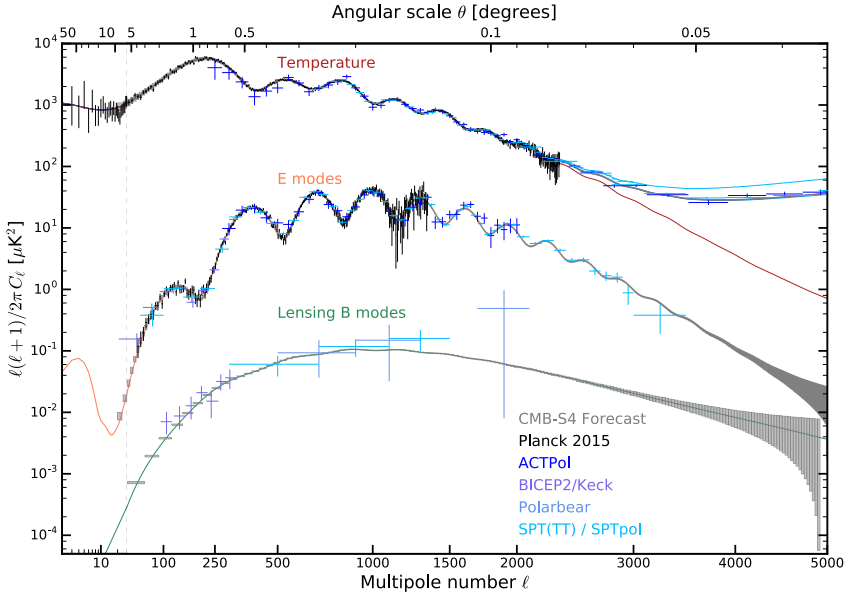


Figure 21: Current status of the measurements of the angular power spectra of the CMB, taken from [1].

5.3 Recent and future experiments

The field of CMB polarization measurements is extremely crowded and rapidly growing. It is hard to provide a comprehensive review of all the ongoing and future experiments; we address the interested reader to the Legacy Archive for Microwave Background Data Analysis (LAMBDA, <https://lambda.gsfc.nasa.gov/>) website, which contains a list²⁴ of the CMB experiments which is updated regularly.

We group the current CMB experiments according to the kind of site chosen for their operation, as this allows us to underline their similarities and better explain their differences:

High-altitude sites: The simplest solution is to deploy instruments at locations where the contribution of atmospheric emission in the frequency range of interest (10–1000 GHz) is low enough not to prevent the measurement (Fig. 22). Only a fraction of the 4π sky can be mea-

²⁴<https://lambda.gsfc.nasa.gov/product/expt/>

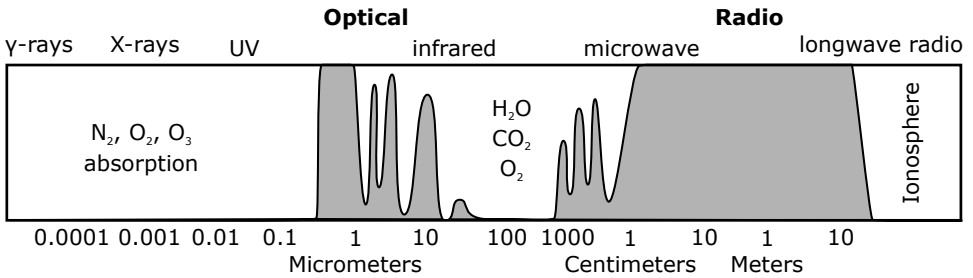


Figure 22: Transparency of atmosphere as a function of the wavelength of electromagnetic radiation. The radio range is limited by the ionosphere at wavelengths greater than a few meters. Atmospheric absorption becomes significant in the sub-millimeter range. The wavelengths that are most interesting for CMB studies are in the range $1000\ \mu\text{m}$ – $1\ \text{cm}$. Adapted from [19].

sured from one site (typically this fraction is no more than 60%). The integration time can be made arbitrarily large, which helps in improving the sensitivity. There are four frequency windows where this is possible [60]: below 15 GHz, 30–50 GHz, 75–110 GHz, and 130–170 GHz (the latter three correspond to the three peaks near 1 cm in Fig. 22). Typically, one wants high-altitude sites with low water vapour pressure and very good weather conditions during the whole year. Some of the most notable used locations are the Atacama Desert in Chile (ACTPol [3], Polarbear [50], CCAT-Prime [101], CLASS [25,26], Simons Array [99], ABS [98]), Antarctica (SPTpol [12], BICEP/Keck [16]), Tenerife (QUIJOTE [90], GroundBIRD [74]), Argentina (QUBIC [89]), and the peculiar case of C-BASS [54], whose two telescopes will be deployed to the Owens Valley Radio Observatory (California) and to South Africa. Experiments tend to be deployed to the same locations, in order to take advantage of existing infrastructures (roads, electric cables, etc.).

Space: This is the most obvious solution if one aims to be completely free of atmospheric contamination and wants to achieve a 100% coverage of the sky sphere. The obvious disadvantages are the costs and the complexity. So far there have been four CMB experiments from space: RELIKT-1 [55], COBE [66], WMAP [10], and Planck [81]. A fifth one, LiteBIRD [67], has been selected by the Japanese Aerospace

eXploration Agency (JAXA) in May, 2019, and it is currently the only space-based CMB mission in preparation; see below for more details.

Stratospheric balloons: In order to reduce the effect of atmospheric contamination and make measurements outside the four atmospheric windows listed above, several experiments have used stratospheric balloons. The main disadvantages of balloons are their limited time (the longest flights last a few weeks), the limit in the mass to be carried, and the difficulty of retrieving the data at the end of the flight. On the other hand, they can have a view of the sky without atmospheric disturbance, much like the one achievable from space, but they typically cost 1/100 of a space mission [30–32]. Notable examples are EBEX [38], Spider [100], PILOT [77], PIPER [78].

Among the most promising next-generation experiments, we spend a few words about two of them:

CMB-S4: The United States assembled an impressive amount of resources and people for their planned ground-based experiment, named “Stage-4” (S4) [1,2], which was unfortunately abruptly shut down in 2025 by the US Government. Nevertheless, we provide a short description, because this experiment is a good example of the kind of experiment that would be needed to make a deep measurement of the CMB polarization from the ground. Two locations were selected: the South Pole would have hosted the telescopes used to perform a deep²⁵, small-scale survey of the sky, and the high Chilean Atacama plateau would have hosted the telescopes for a large-scale survey. As stated in the CMB-S4 Science Book [1], *CMB-S4 is intended to be the definitive ground-based experiment*. Its aim was to reach the cosmic variance limit (Eq. 16) for all the angular scales accessible from ground, which means all but the lowest ones ($\ell \lesssim 10$). There would have been many telescopes, and a few of them would have reached angular resolutions smaller than 5′: this is needed for removing foreground contaminations like lensing (Sect. 4.4).

LiteBIRD: This is a space mission presented to the Japan Aerospace Exploration Agency (JAXA) [67]. Its purpose is to detect B-mode polarization with an uncertainty in the tensor-to-scalar-ratio r of the

²⁵A *deep* survey aims to measure the signal from a small patch on the sky with high signal-to-noise ratio.

order of 10^{-3} . The kind of technology used is still under discussion: potential solutions are TES-based bolometers or KID devices (see Sect. 5.1). The frequency coverage is 40–400 GHz, to be achieved with two separate instruments mounted onboard the spacecraft. The Phase-A phase of the design was completed successfully, and the mission has been selected by JAXA on May, 21st 2019. The mission underwent a redesign in 2025 to reduce the costs, but the overall design and the scientific target remained the same.

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